

BFS Traversal

Step 1: Start at E

Legend

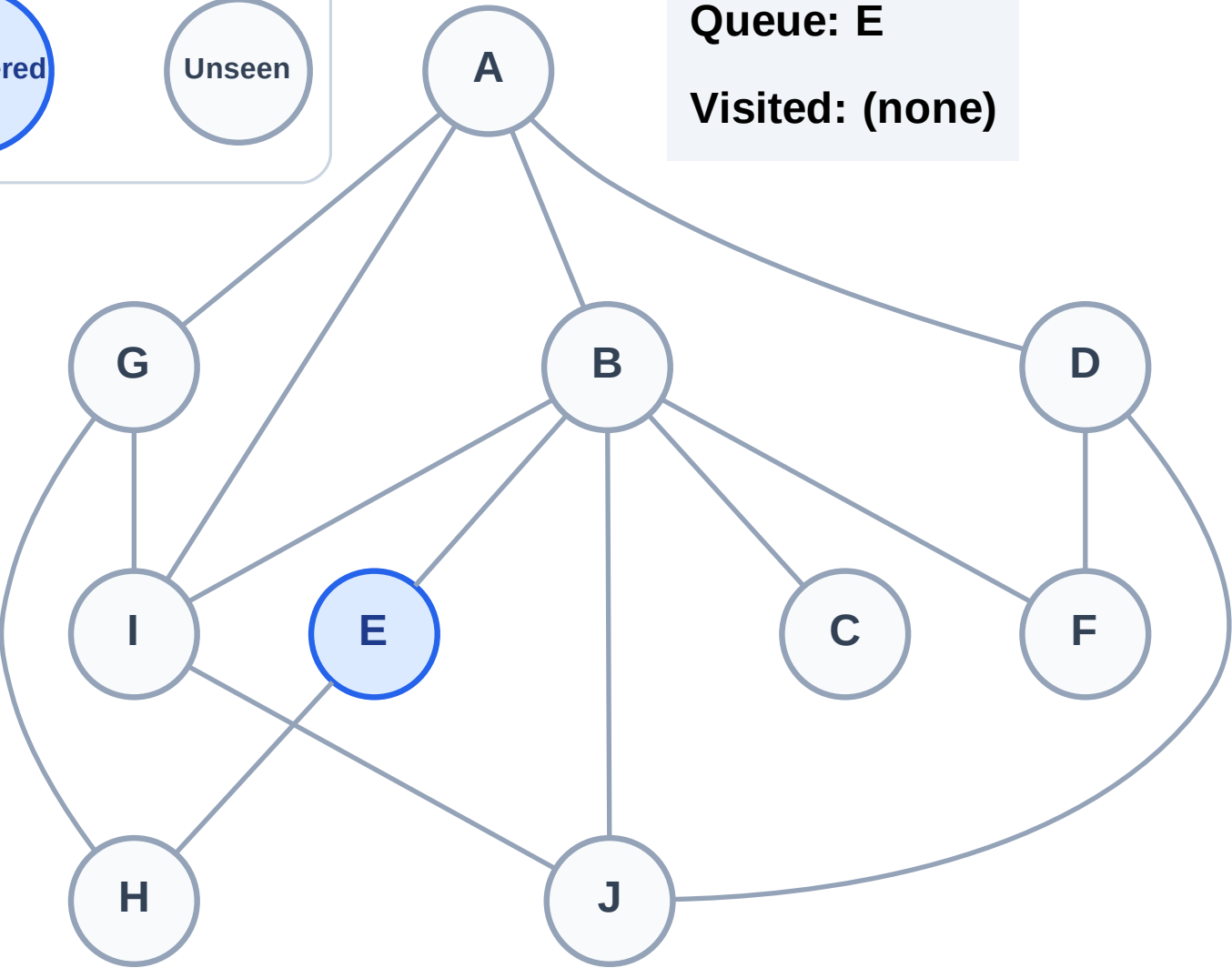
Complete

Processing

Discovered

Unseen

Queue: E
Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

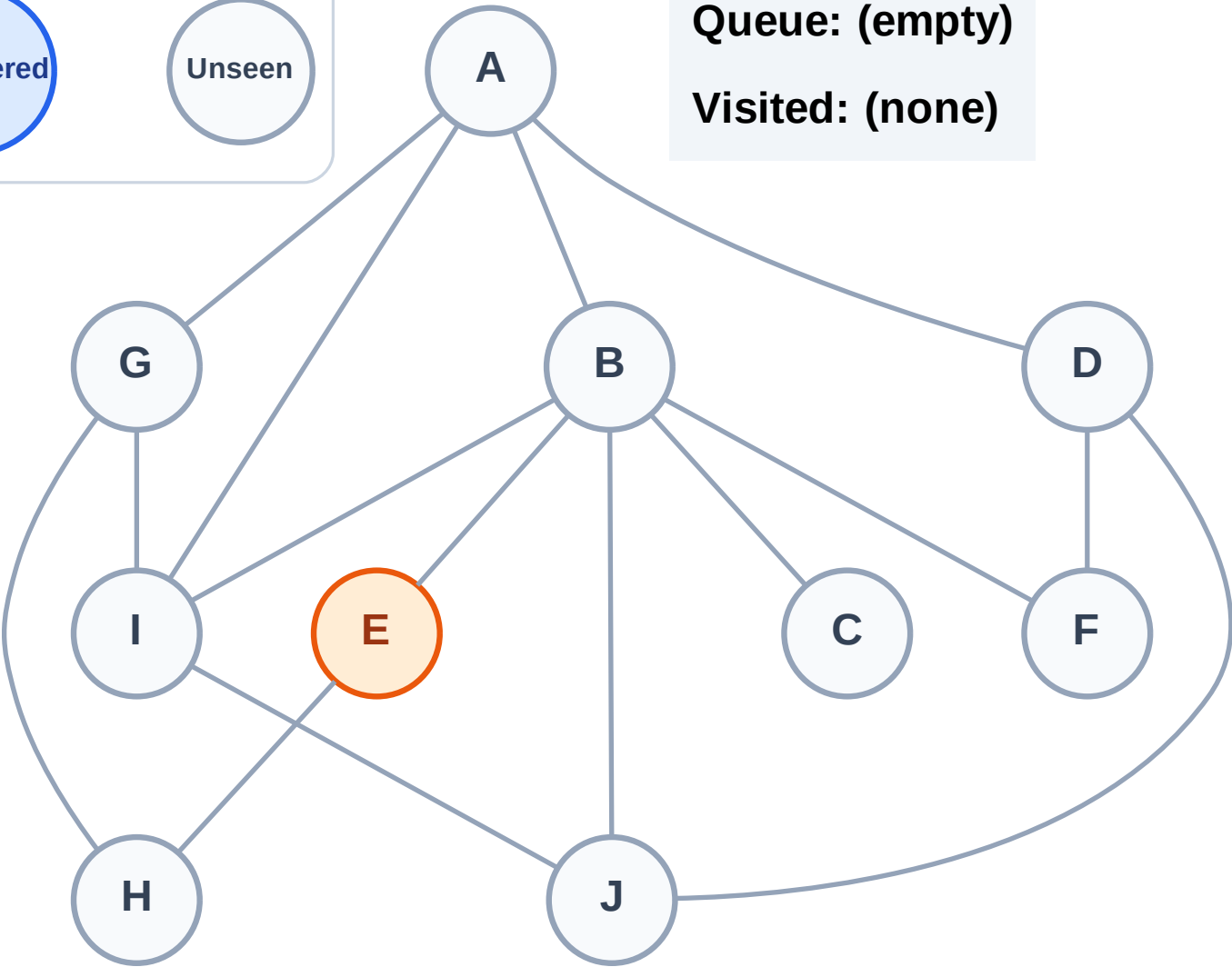
Complete

Processing

Discovered

Unseen

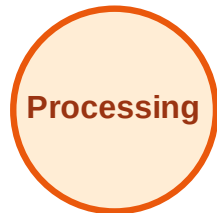
Queue: (empty)
Visited: (none)



BFS Traversal

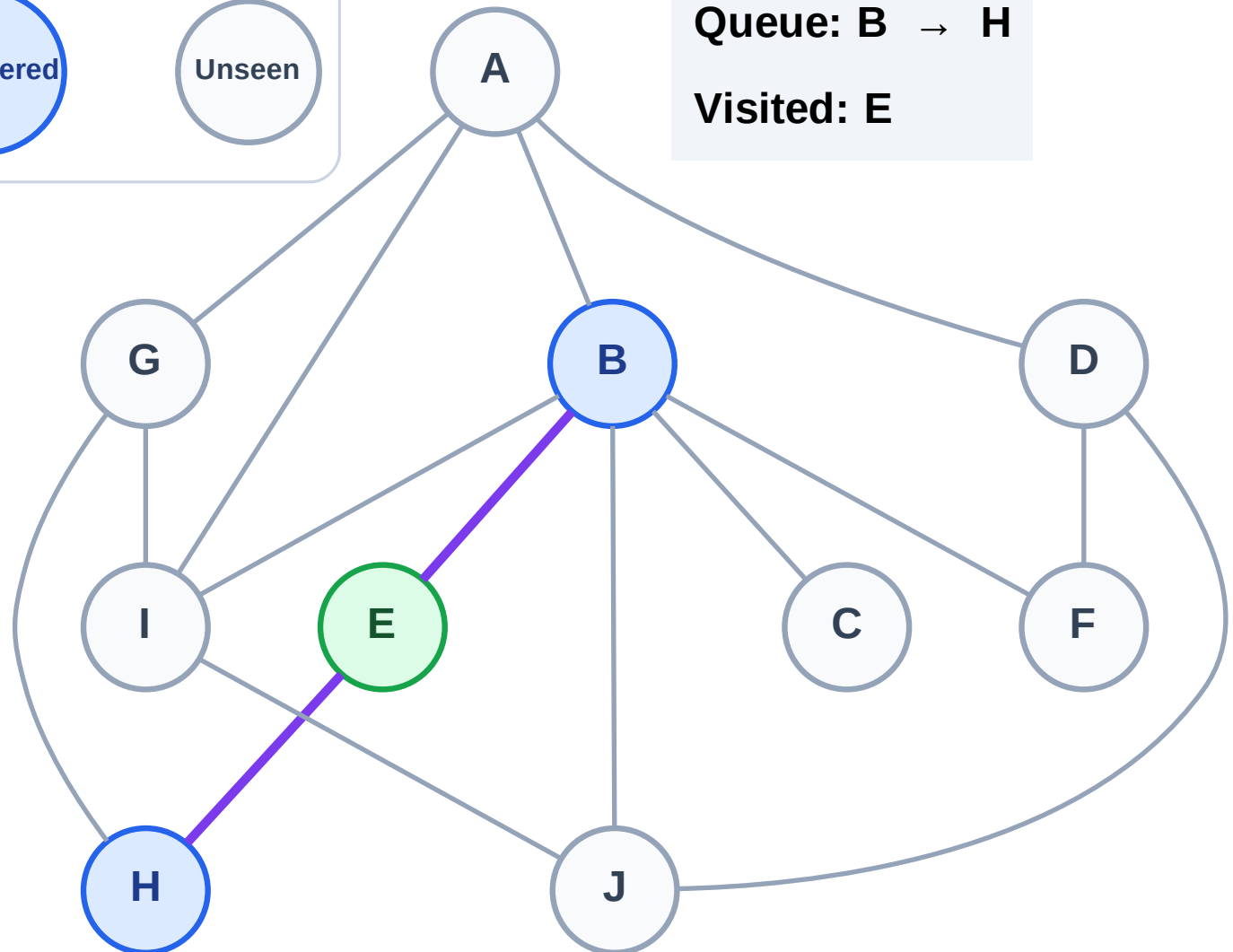
Step 3: Discover B, H from E

Legend



Queue: B → H

Visited: E



BFS Traversal

Step 4: Process node B

Legend

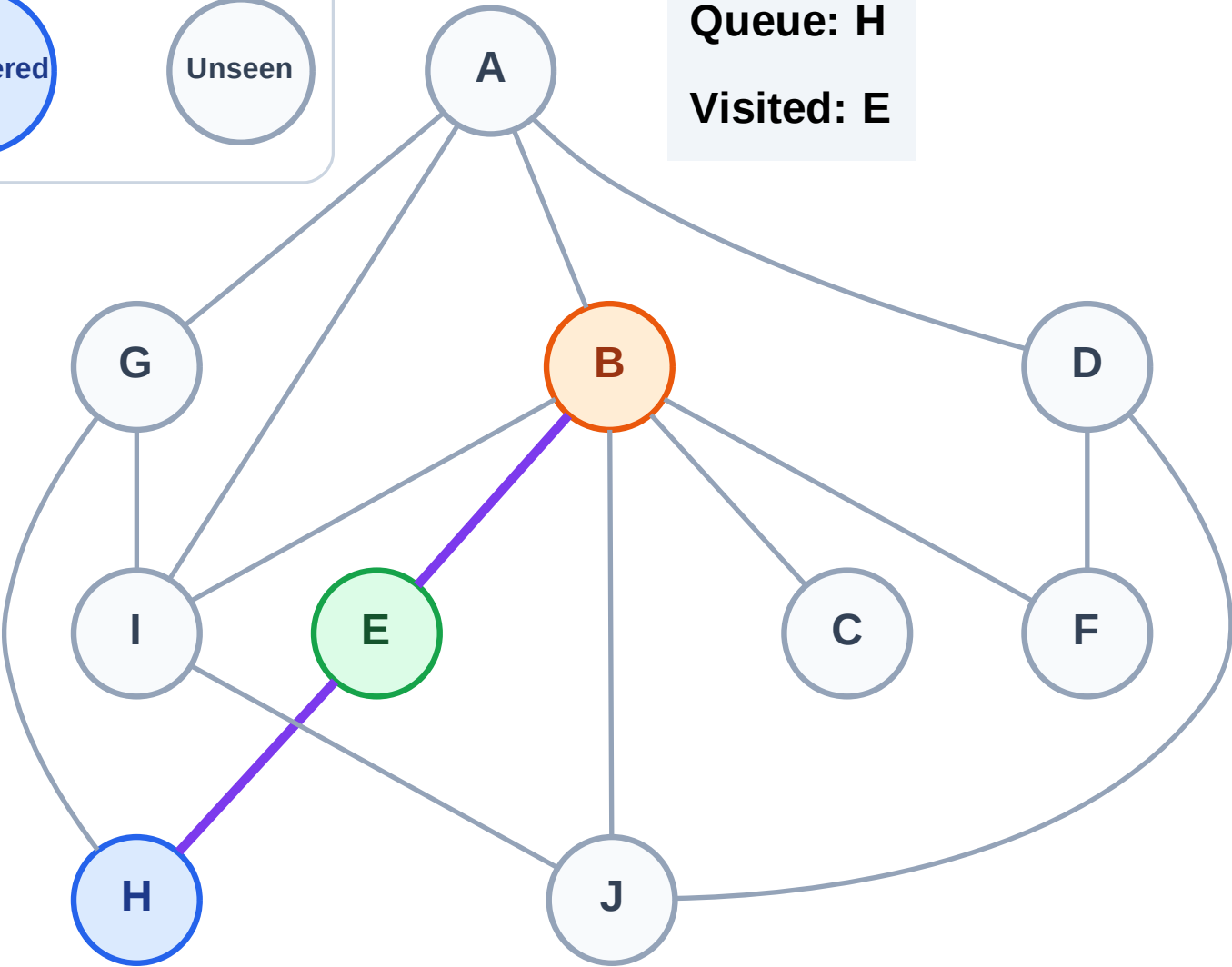
Complete

Processing

Discovered

Unseen

Queue: H
Visited: E



BFS Traversal

Step 5: Discover A, C, F, I, J from B

Legend

Complete

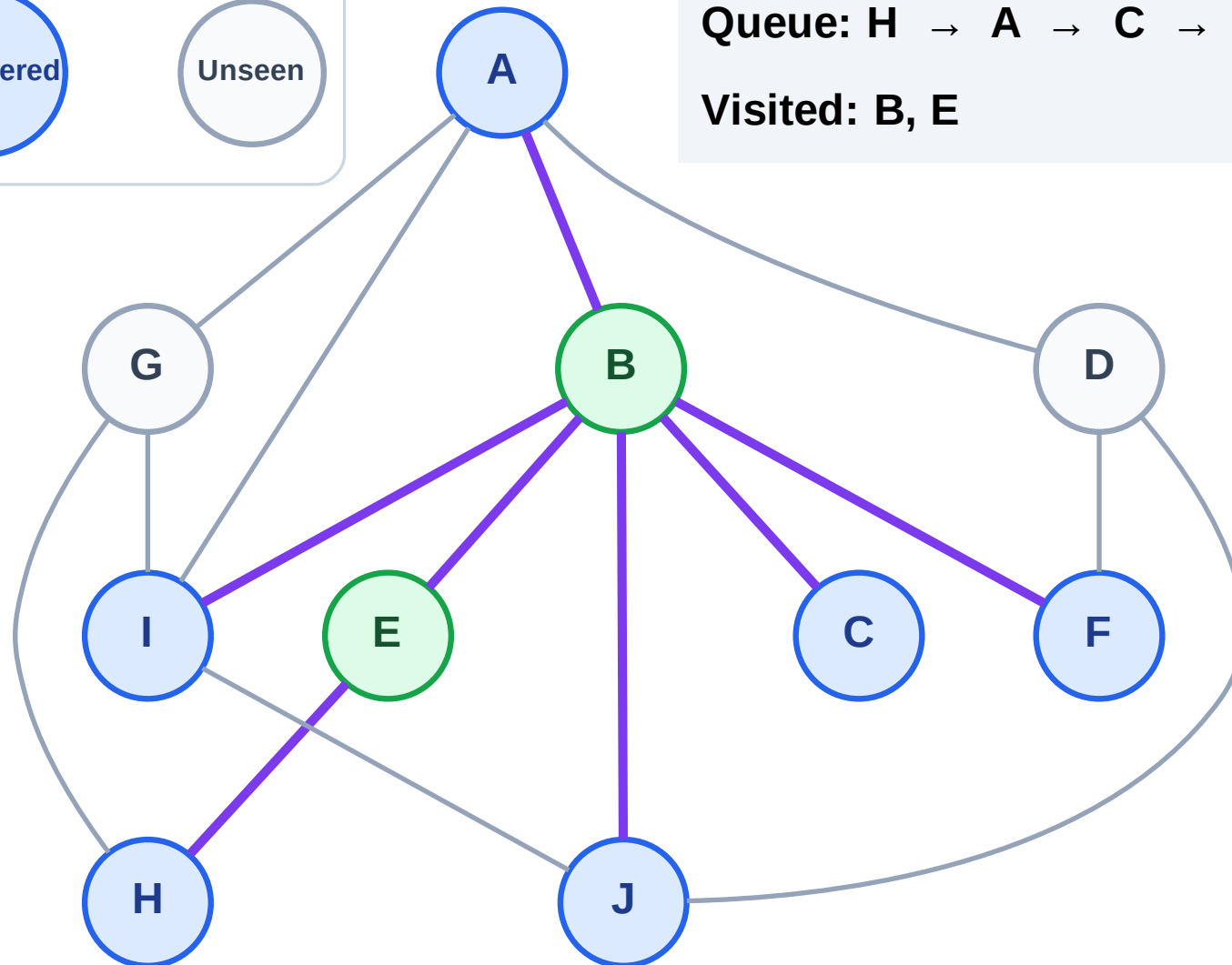
Processing

Discovered

Unseen

Queue: H → A → C → F → I → J

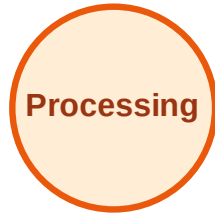
Visited: B, E



BFS Traversal

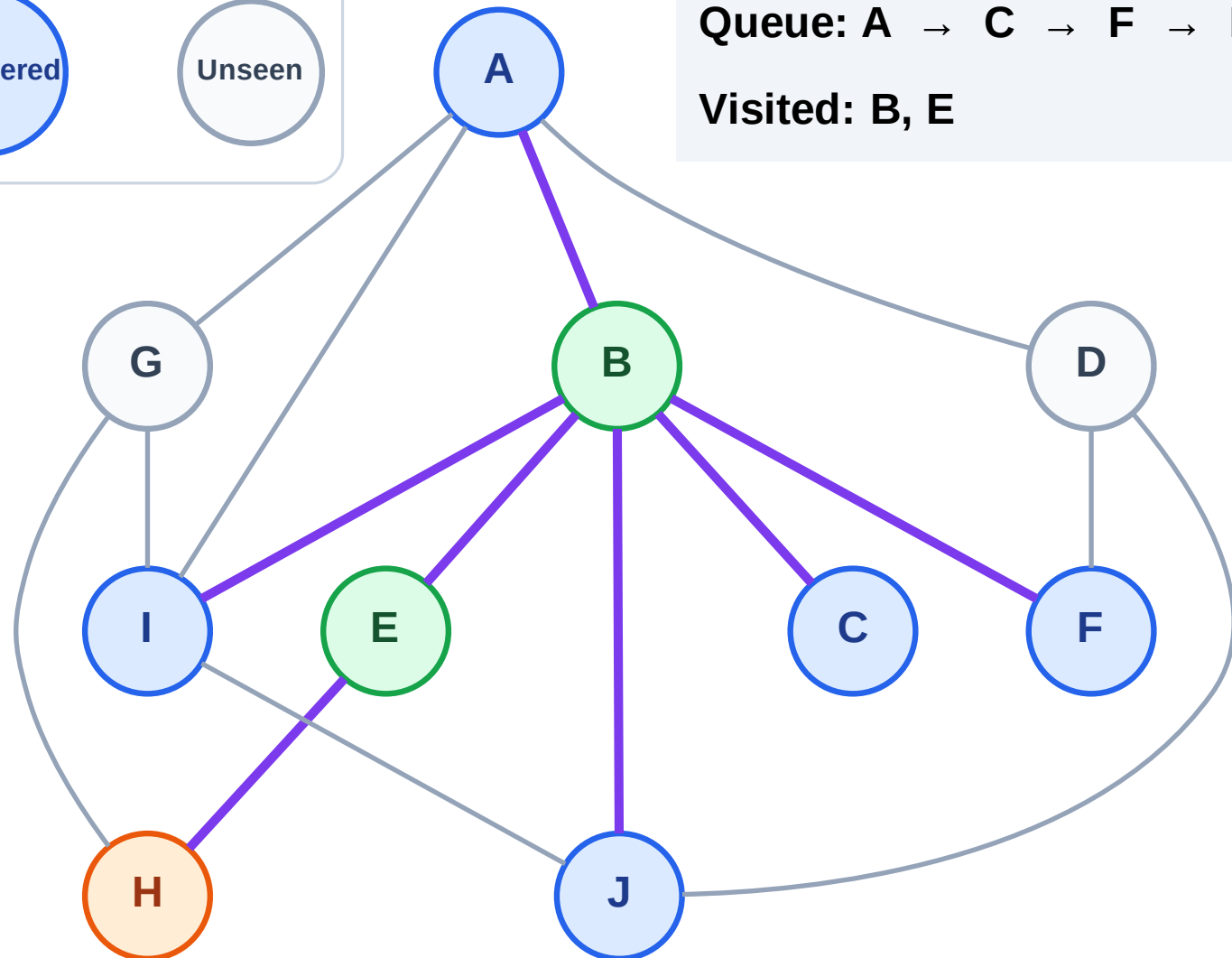
Step 6: Process node H

Legend



Queue: A → C → F → I → J

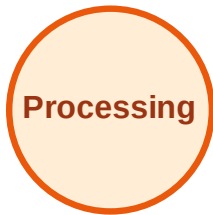
Visited: B, E



BFS Traversal

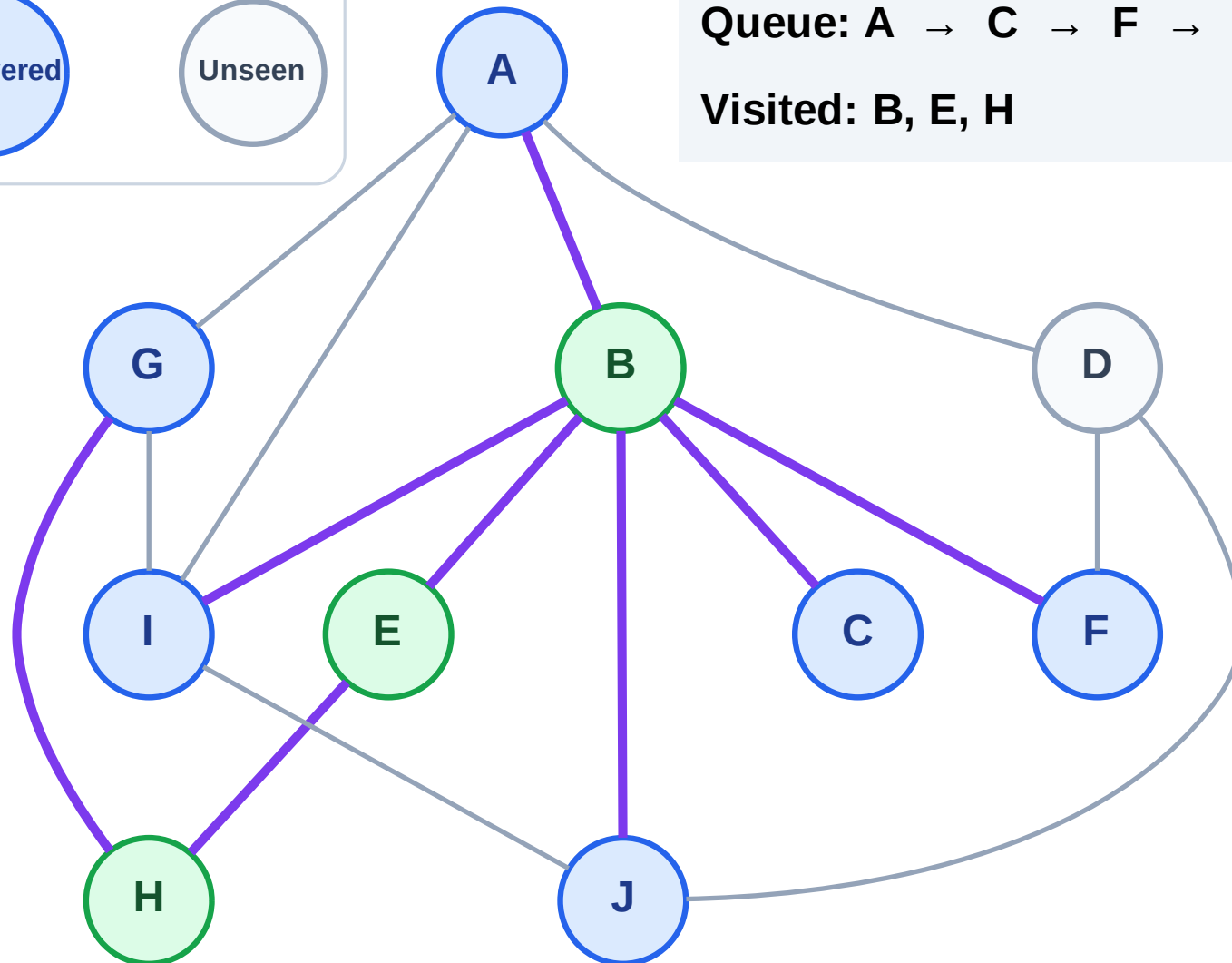
Step 7: Discover G from H

Legend



Queue: A → C → F → I → J → G

Visited: B, E, H



BFS Traversal

Step 8: Process node A

Legend

Complete

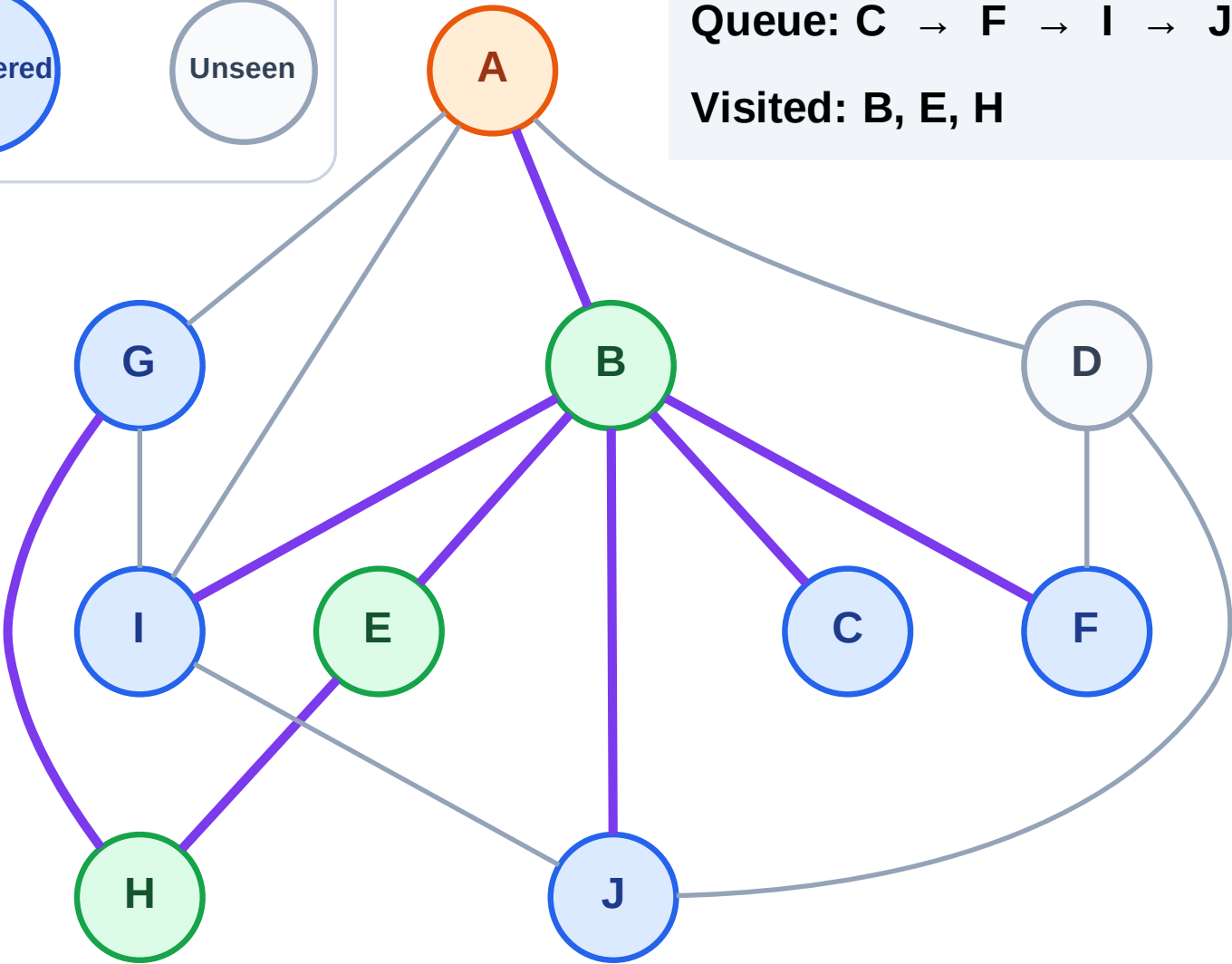
Processing

Discovered

Unseen

Queue: C → F → I → J → G

Visited: B, E, H



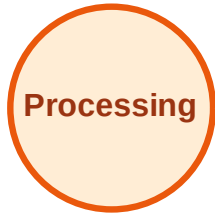
BFS Traversal

Step 9: Discover D from A

Legend



Complete



Processing



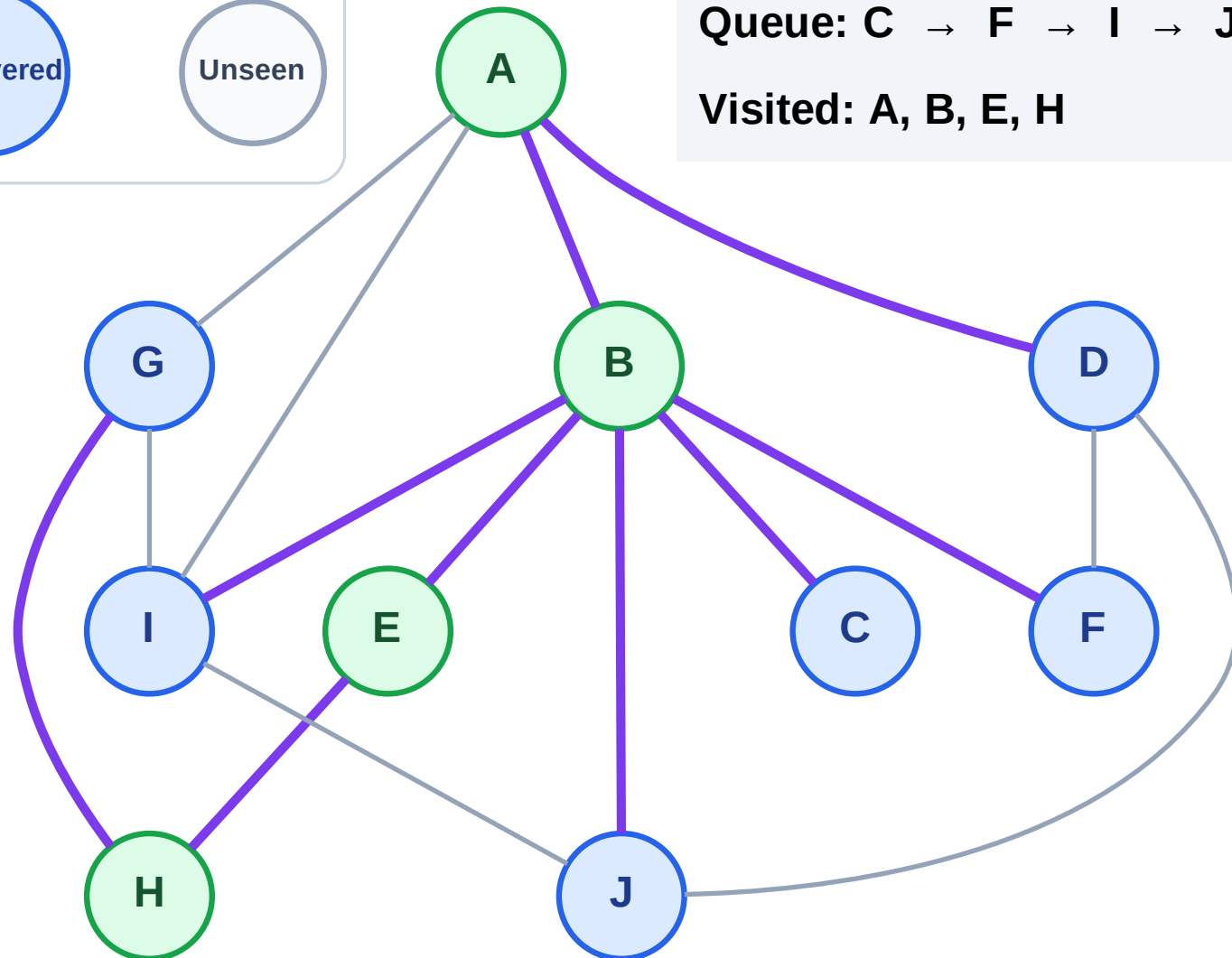
Discovered



Unseen

Queue: C → F → I → J → G → D

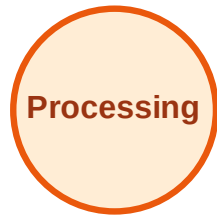
Visited: A, B, E, H



BFS Traversal

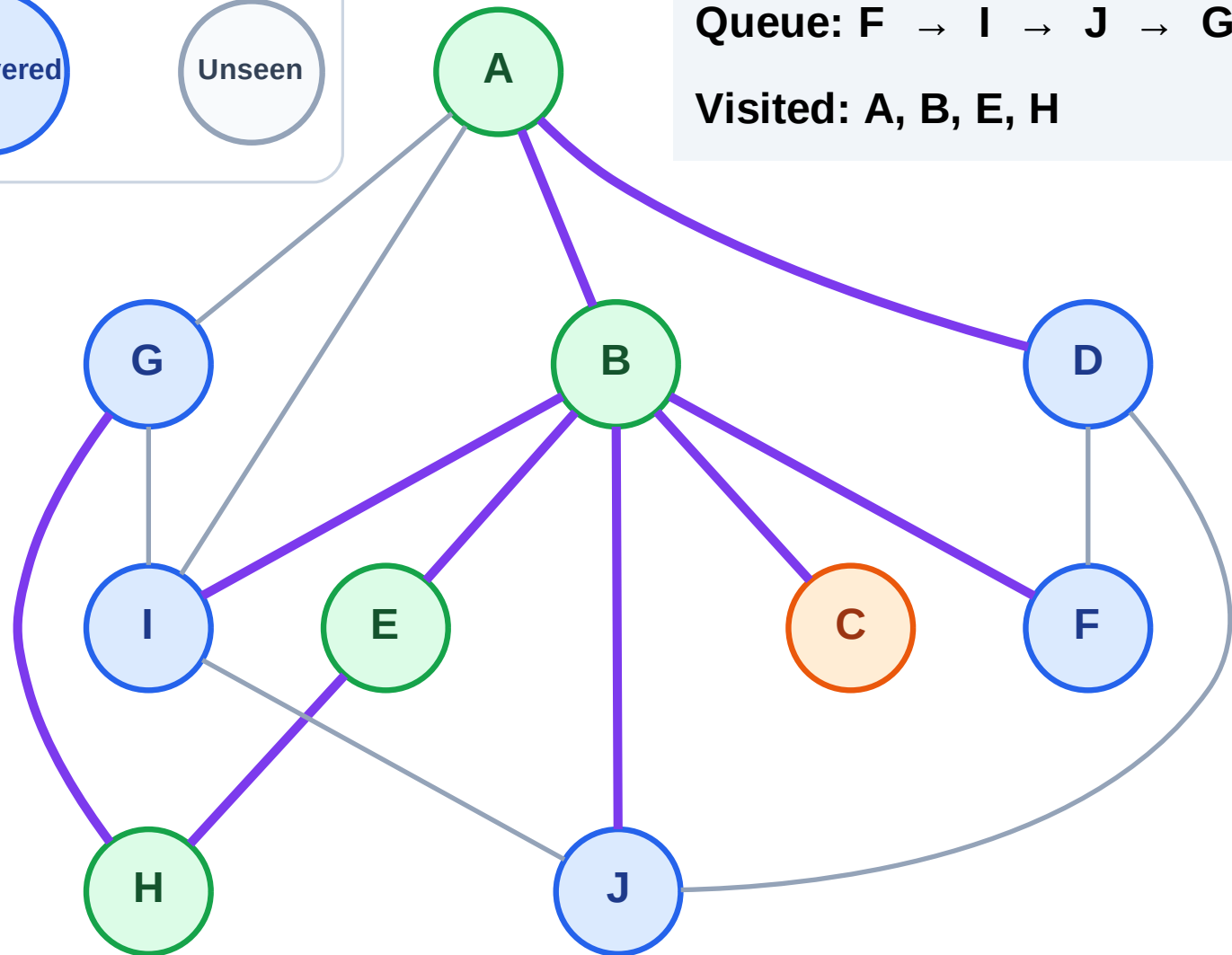
Step 10: Process node C

Legend



Queue: F → I → J → G → D

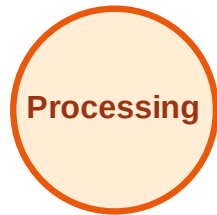
Visited: A, B, E, H



BFS Traversal

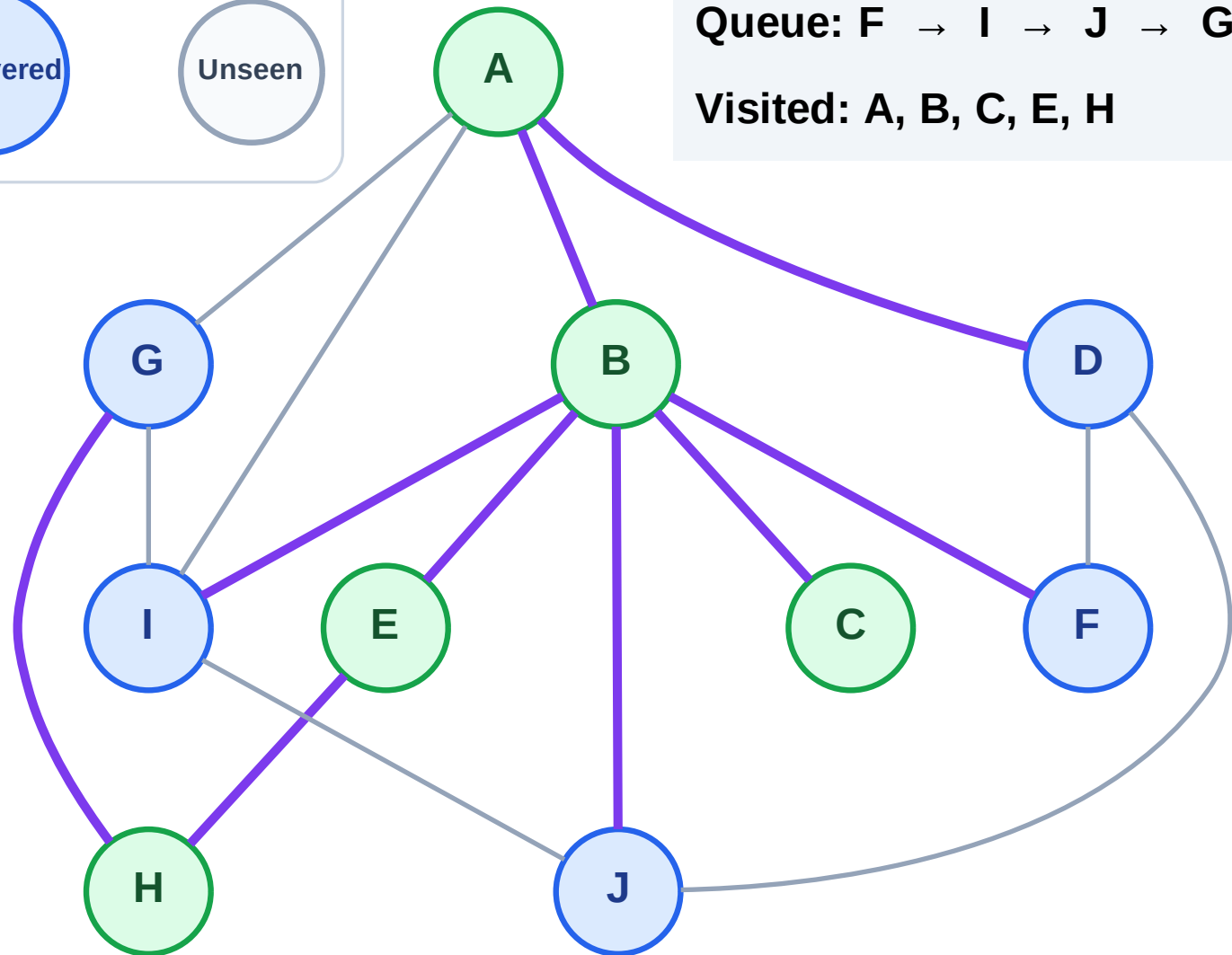
Step 11: No new neighbors from C

Legend



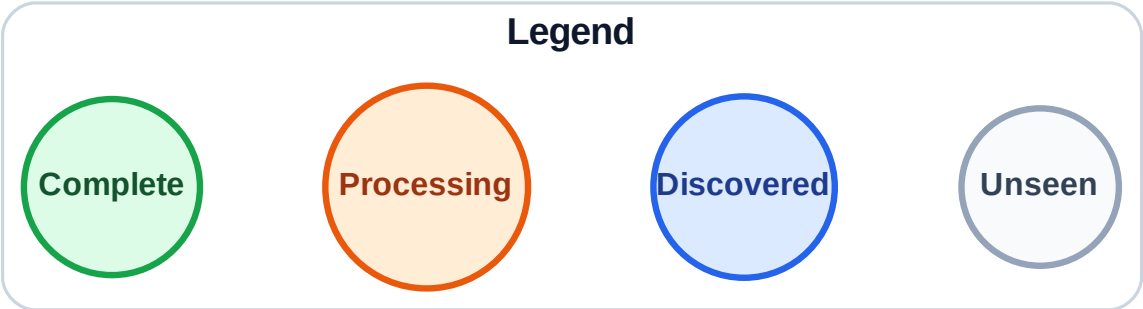
Queue: F → I → J → G → D

Visited: A, B, C, E, H

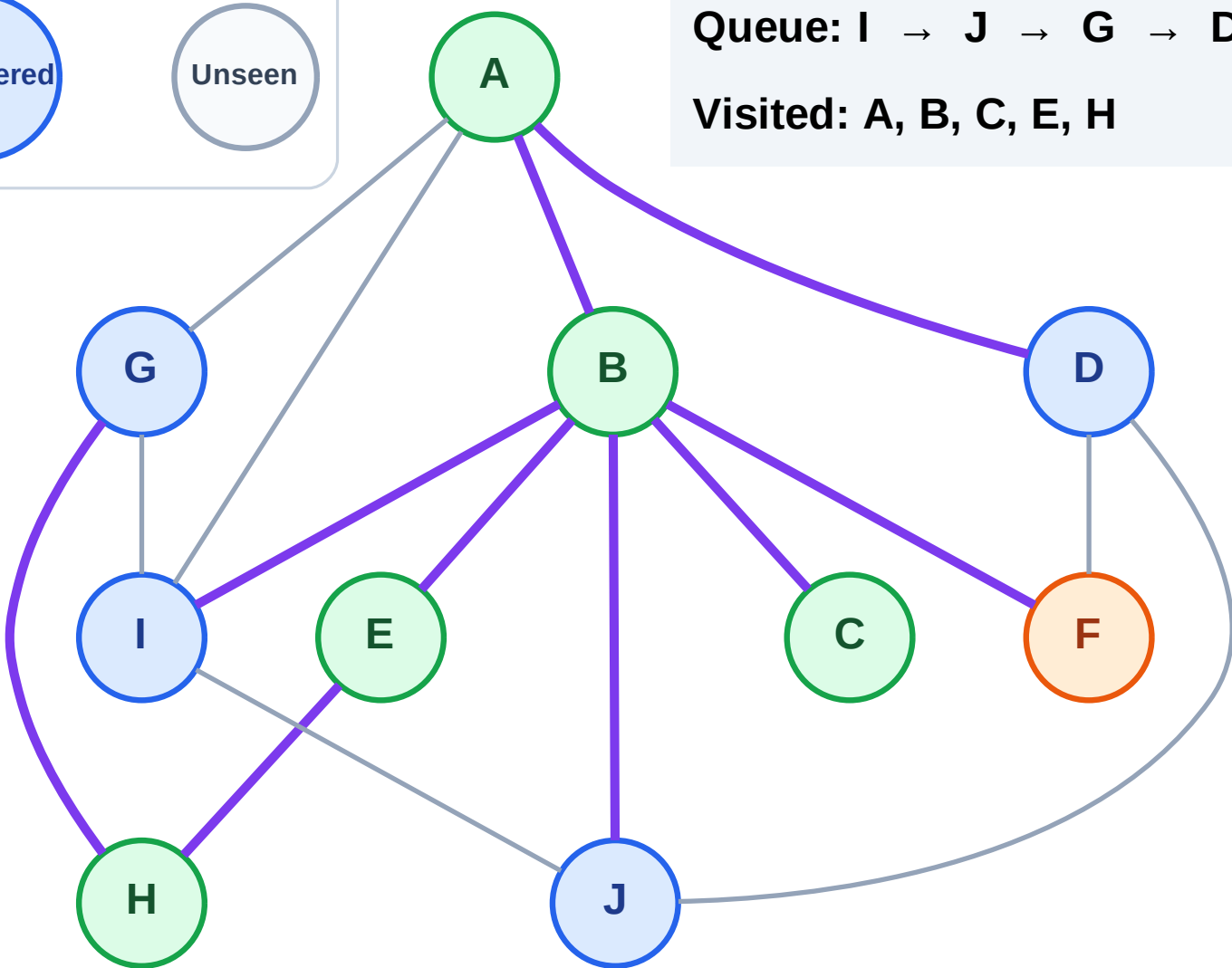


BFS Traversal

Step 12: Process node F



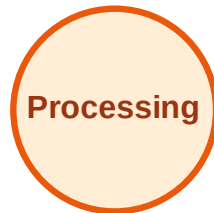
Queue: I → J → G → D
Visited: A, B, C, E, H



BFS Traversal

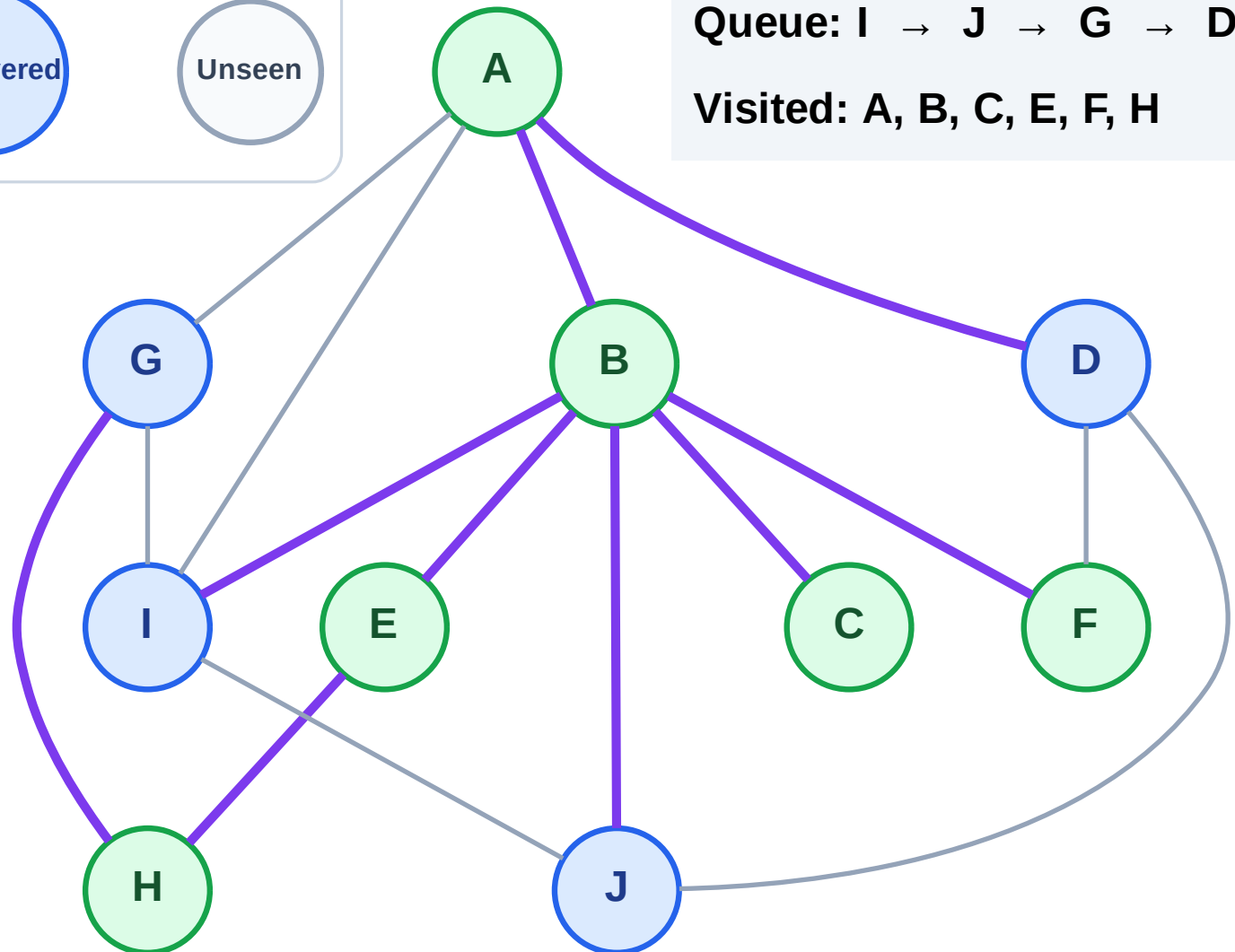
Step 13: No new neighbors from F

Legend



Queue: I → J → G → D

Visited: A, B, C, E, F, H



BFS Traversal

Step 14: Process node I

Legend

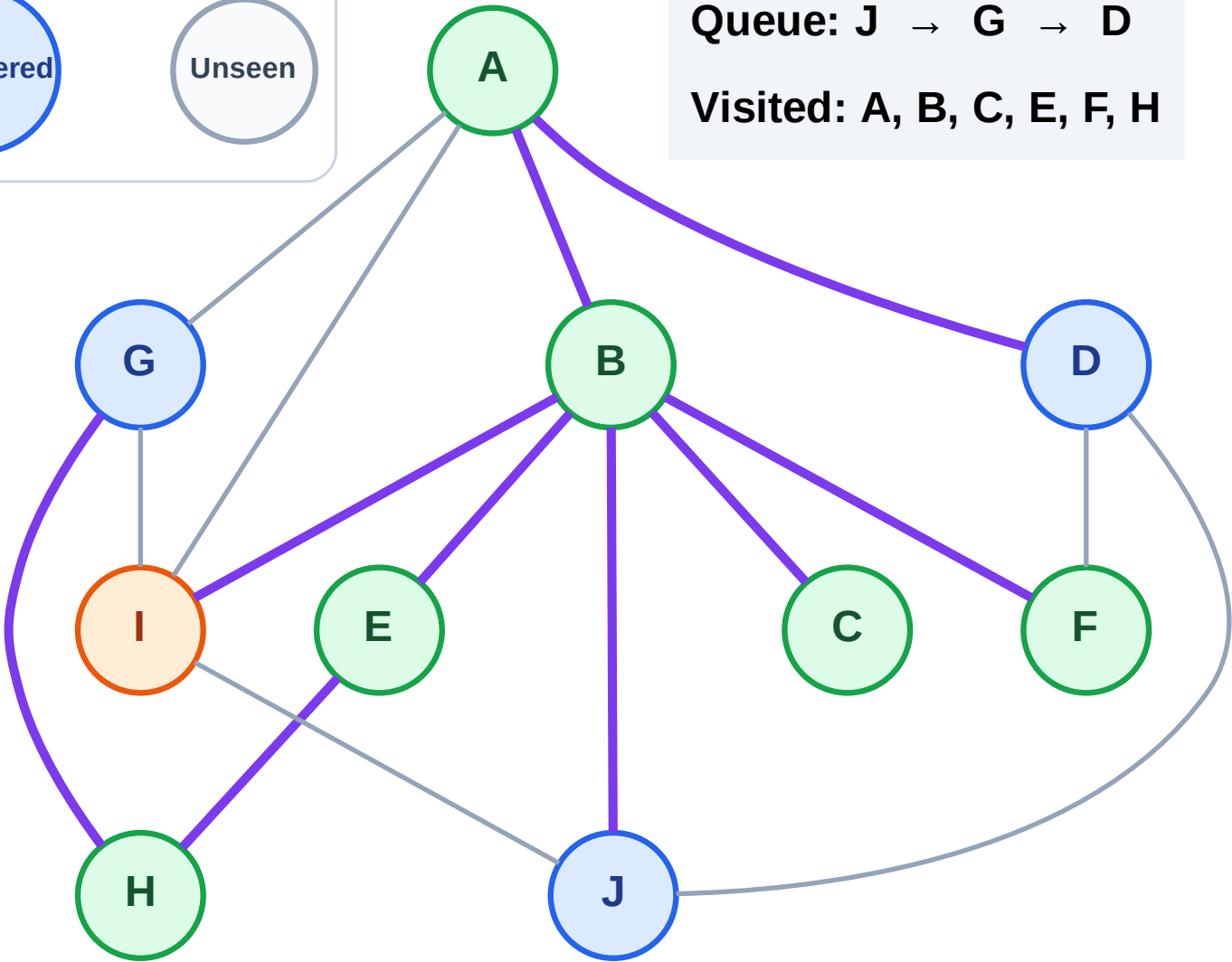
Complete

Processing

Discovered

Unseen

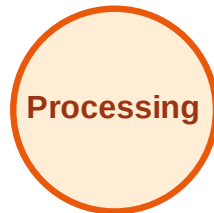
Queue: J → G → D
Visited: A, B, C, E, F, H



BFS Traversal

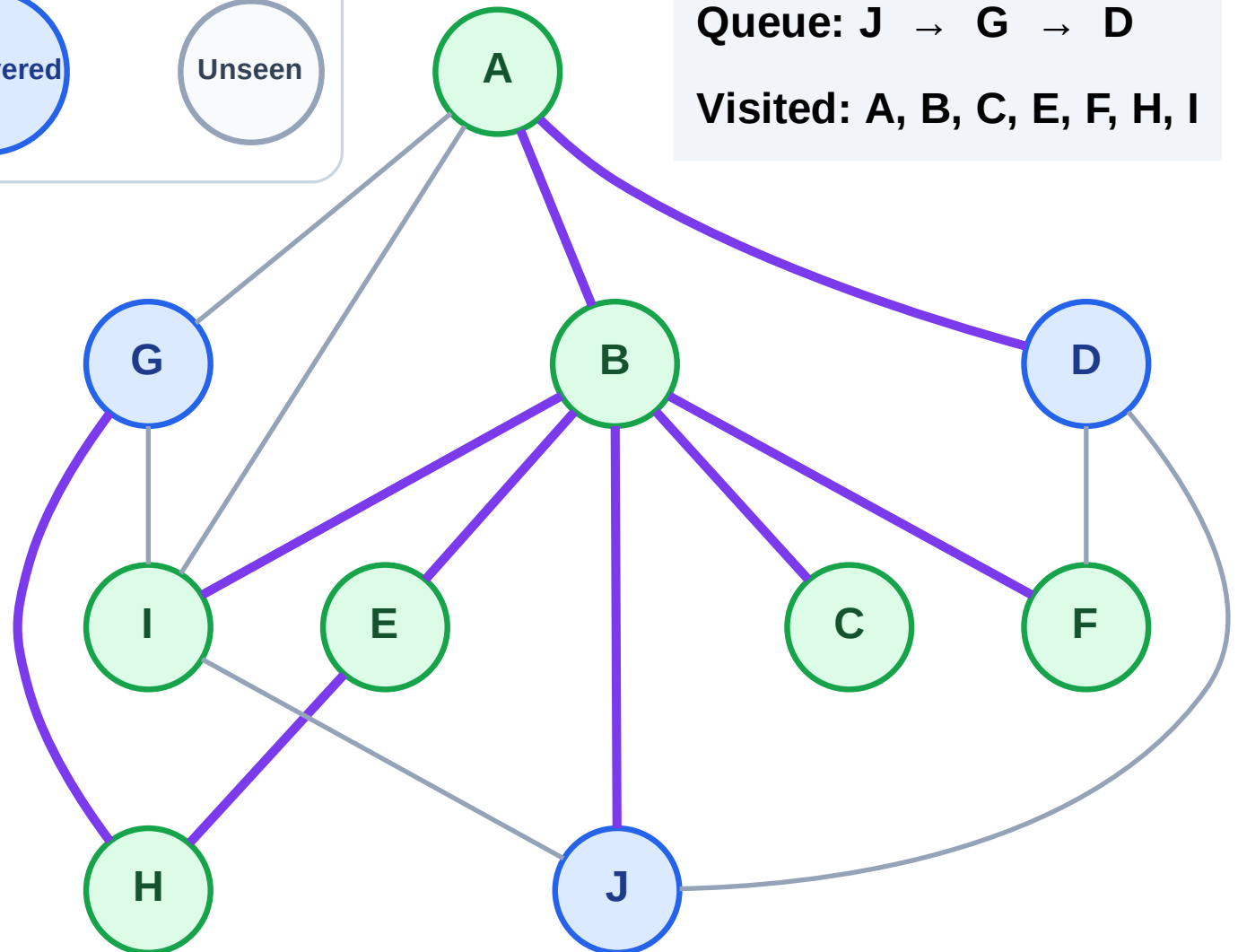
Step 15: No new neighbors from I

Legend



Queue: J → G → D

Visited: A, B, C, E, F, H, I



BFS Traversal

Step 16: Process node J

Complete

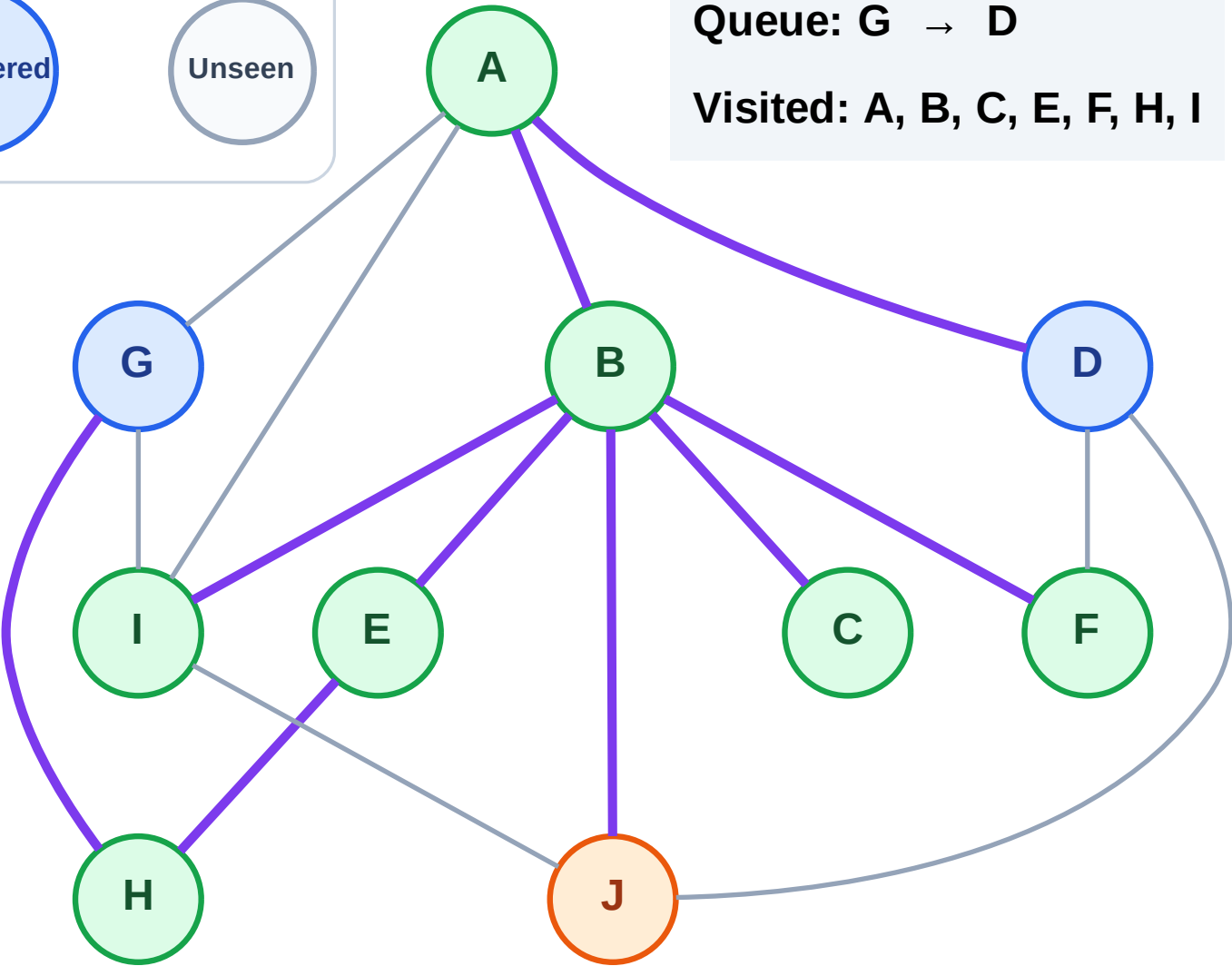
Processing

Discovered

Unseen

Queue: G → D

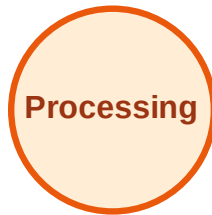
Visited: A, B, C, E, F, H, I



BFS Traversal

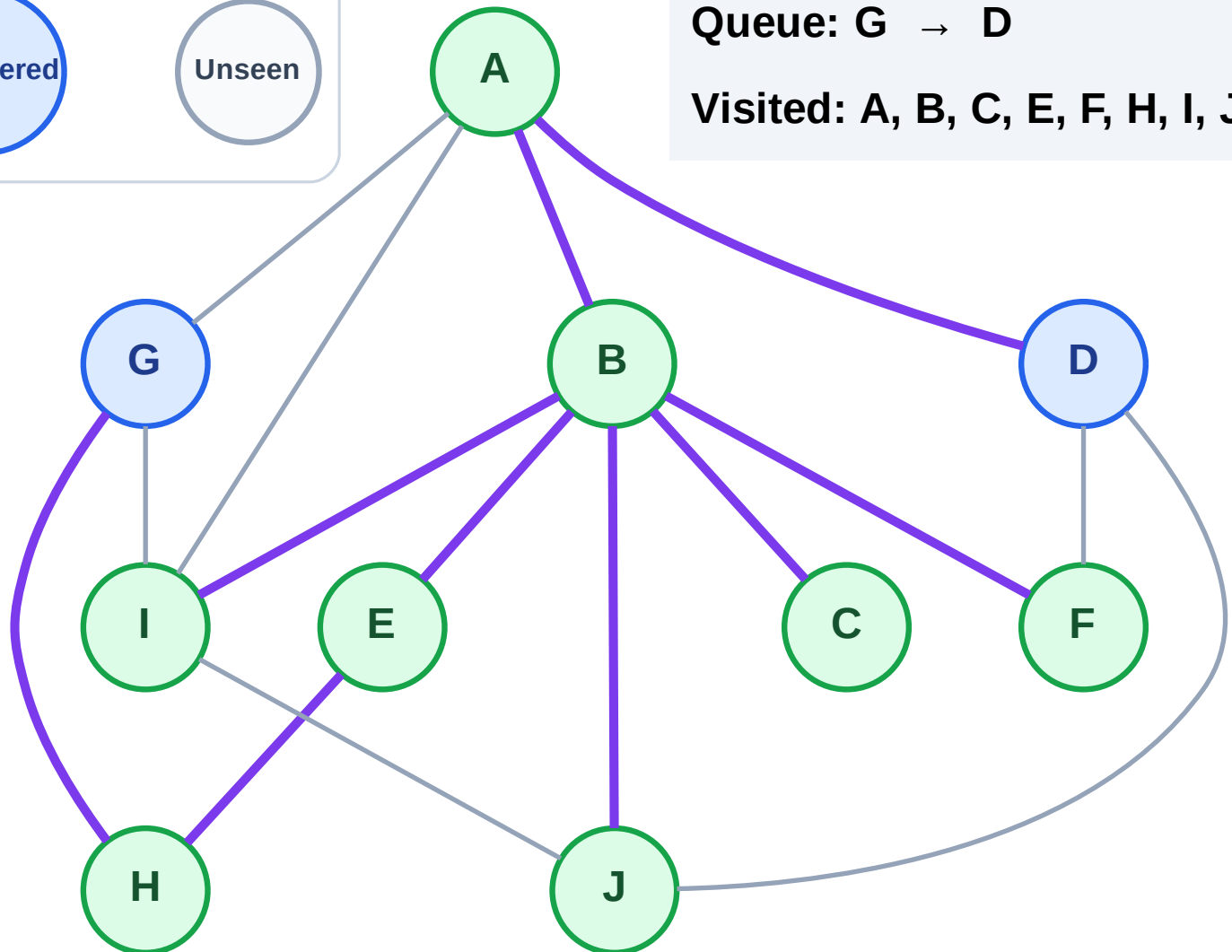
Step 17: No new neighbors from J

Legend



Queue: G → D

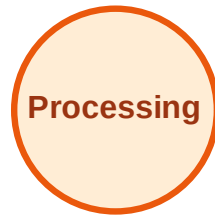
Visited: A, B, C, E, F, H, I, J



BFS Traversal

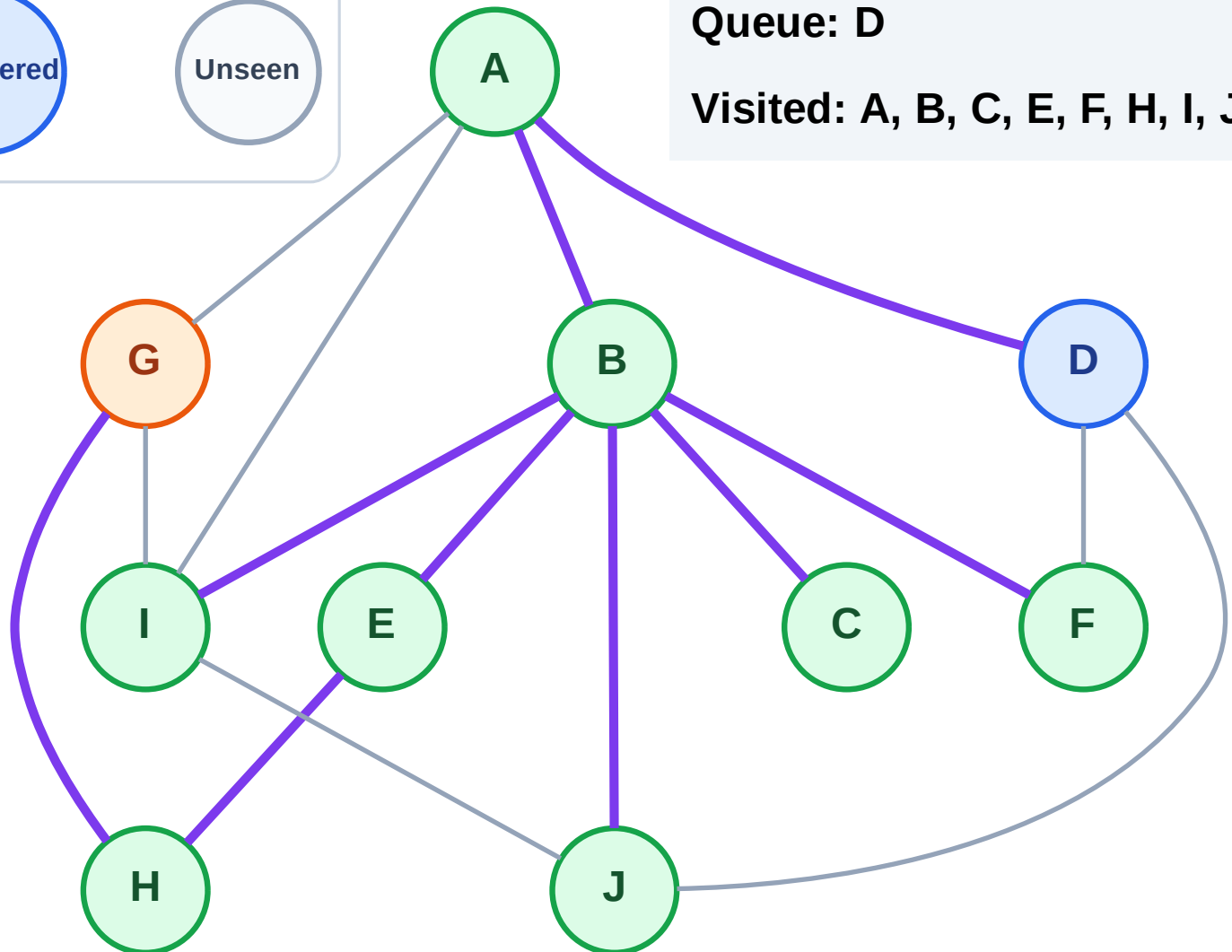
Step 18: Process node G

Legend



Queue: D

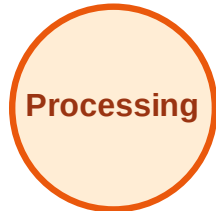
Visited: A, B, C, E, F, H, I, J



BFS Traversal

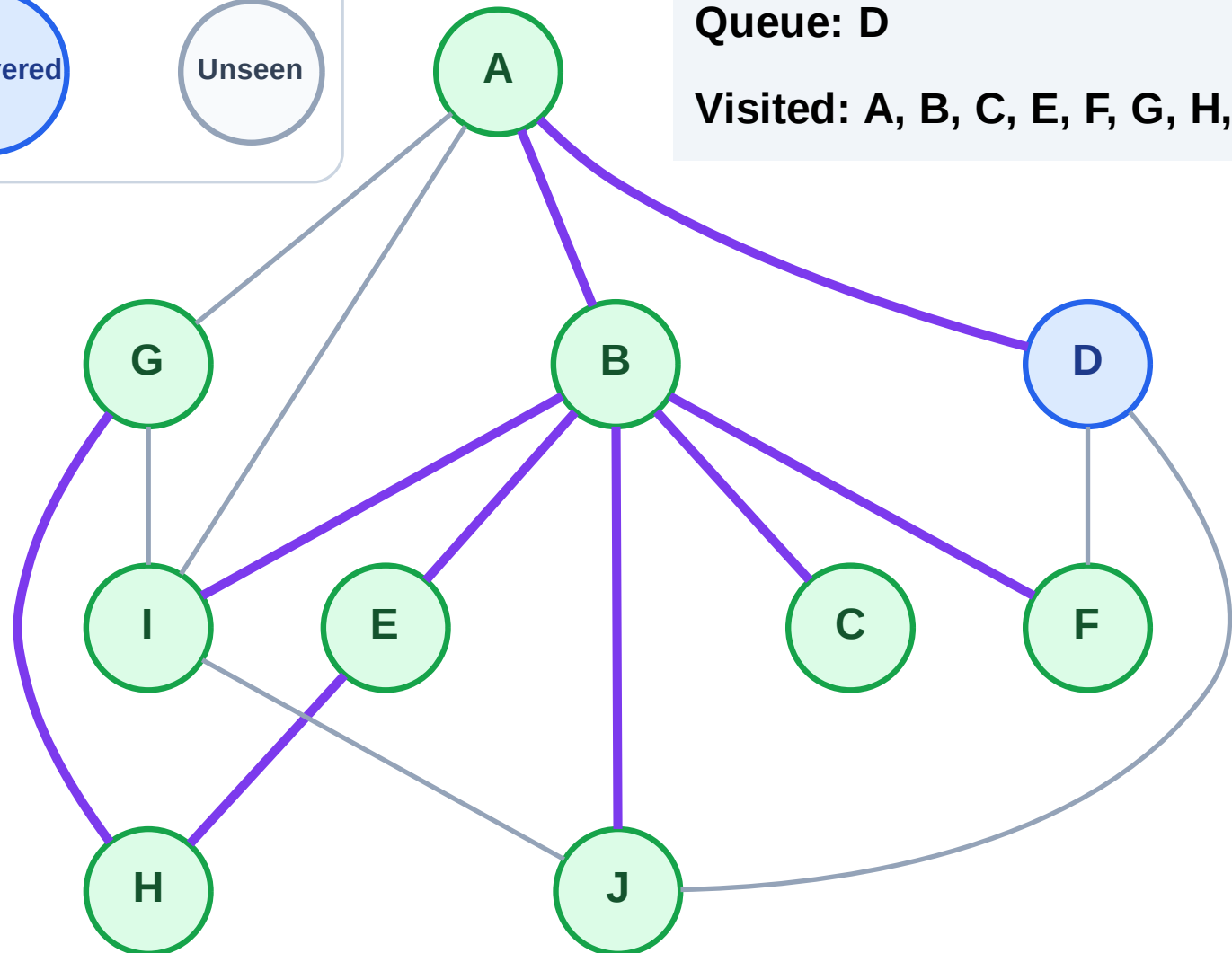
Step 19: No new neighbors from G

Legend



Queue: D

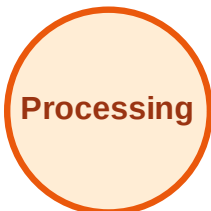
Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

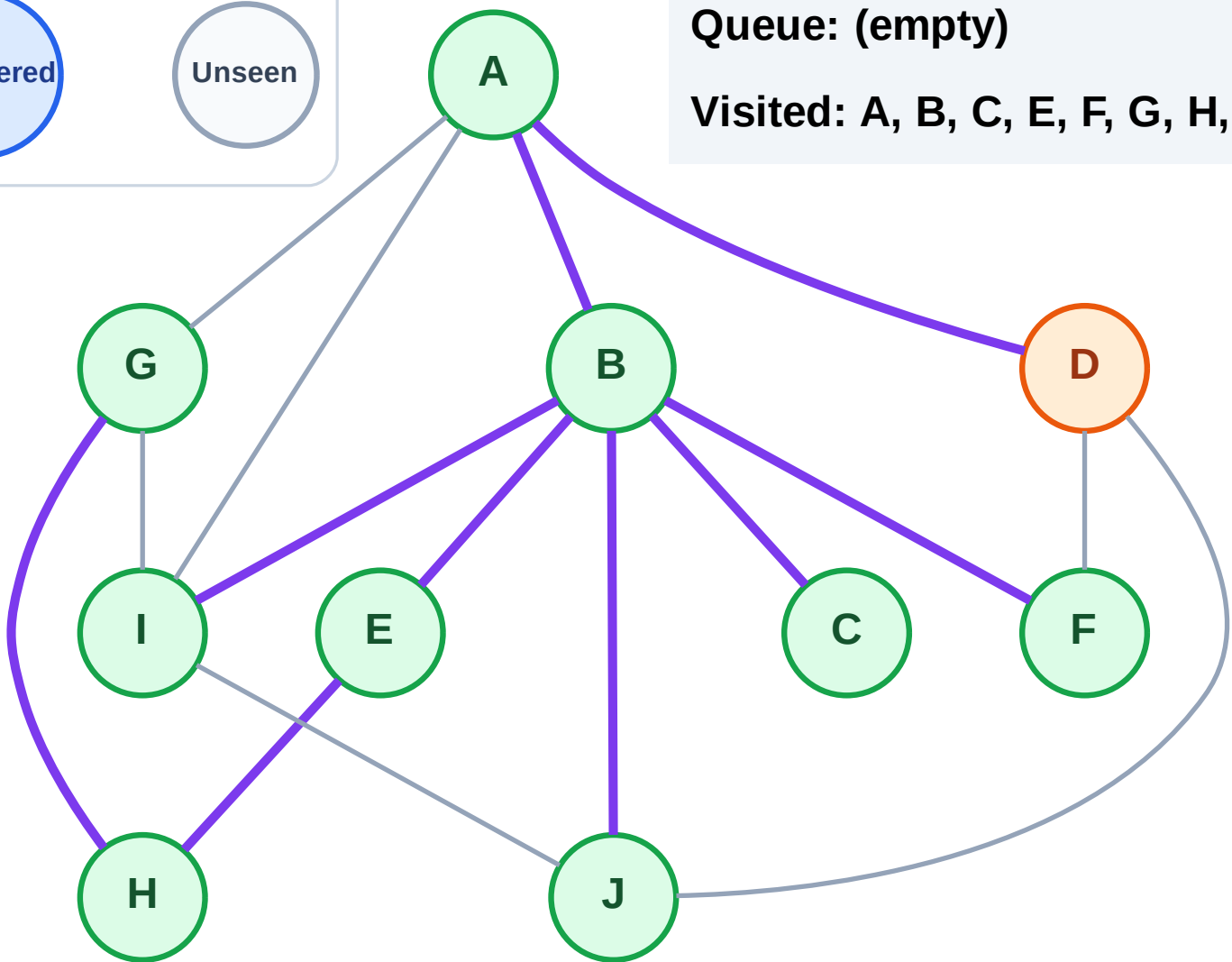
Step 20: Process node D

Legend



Queue: (empty)

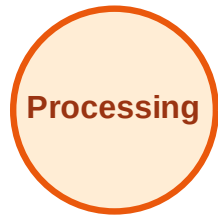
Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

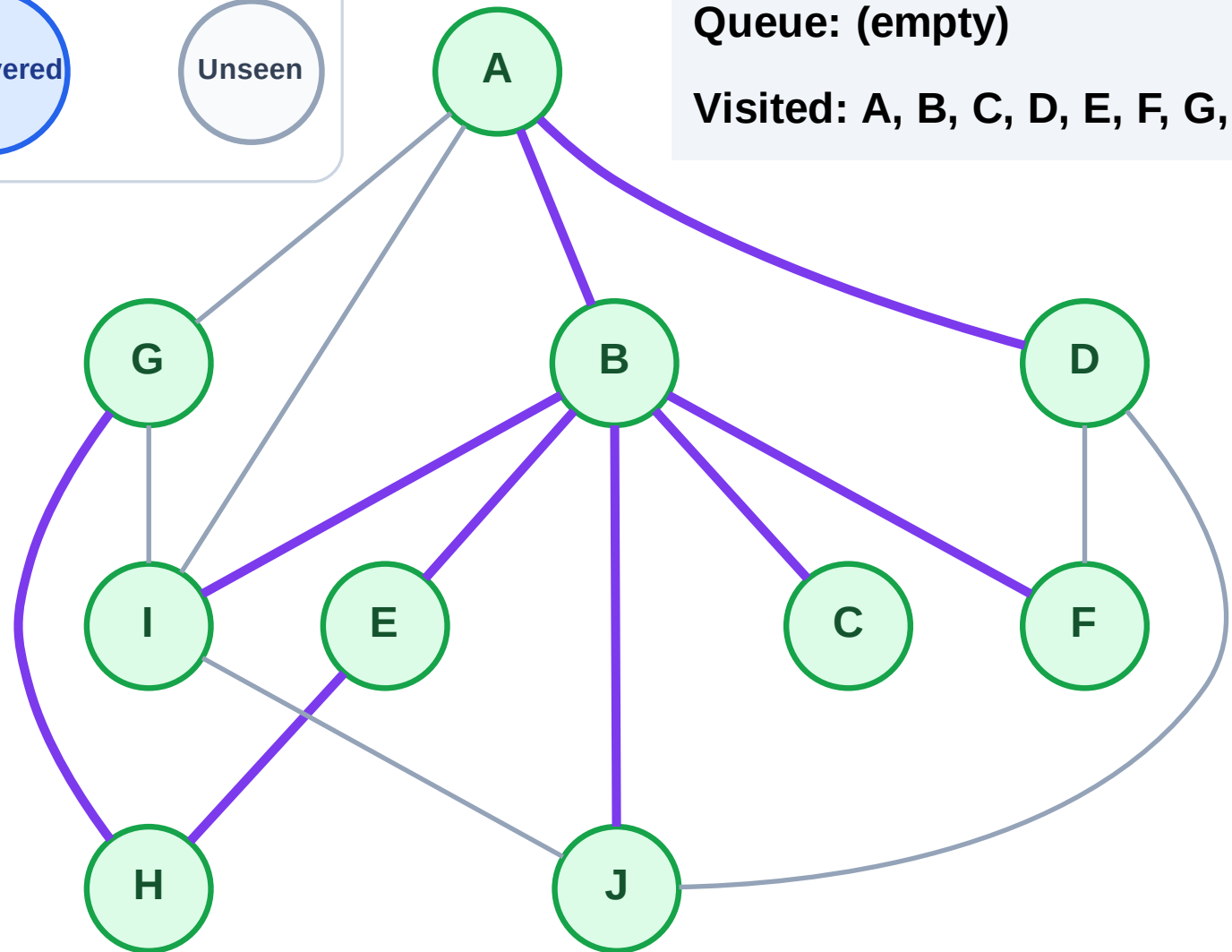
Step 21: No new neighbors from D

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

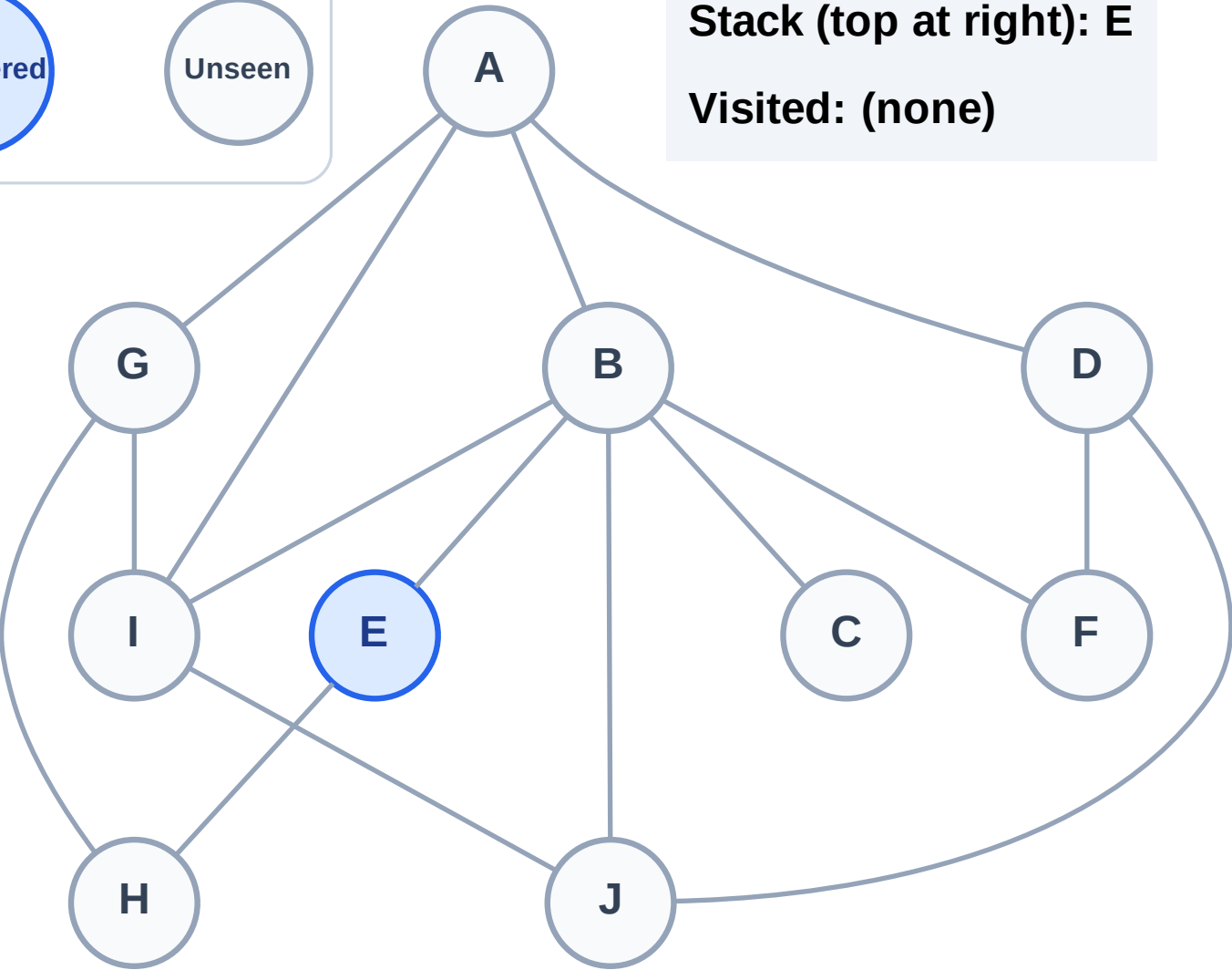
Complete

Processing

Discovered

Unseen

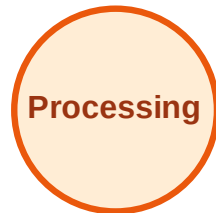
Stack (top at right): E
Visited: (none)



DFS Traversal

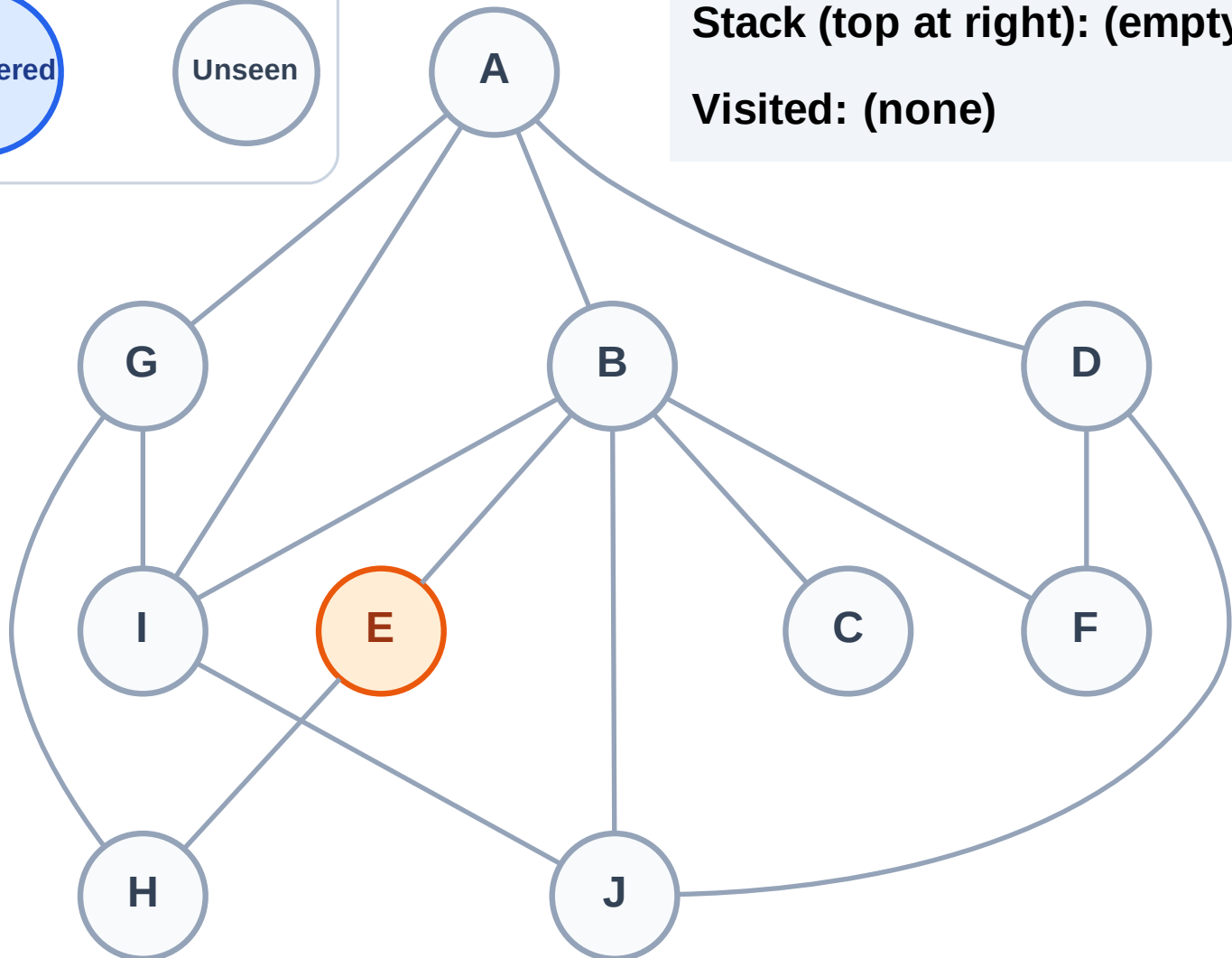
Step 2: Process node E

Legend



Stack (top at right): (empty)

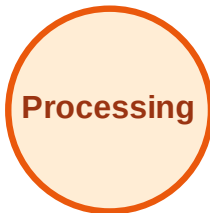
Visited: (none)



DFS Traversal

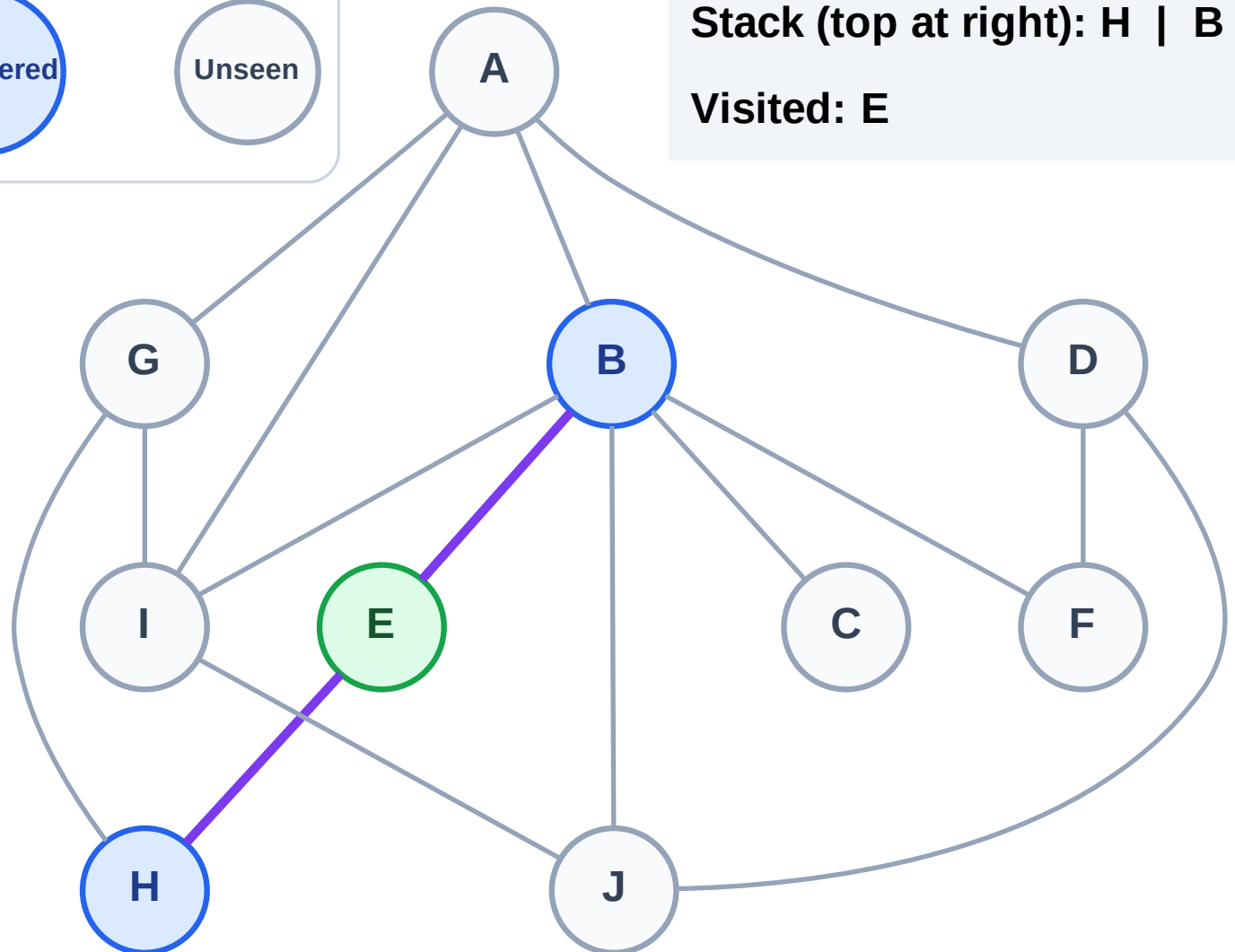
Step 3: Discover H, B from E

Legend



Stack (top at right): H | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

Complete

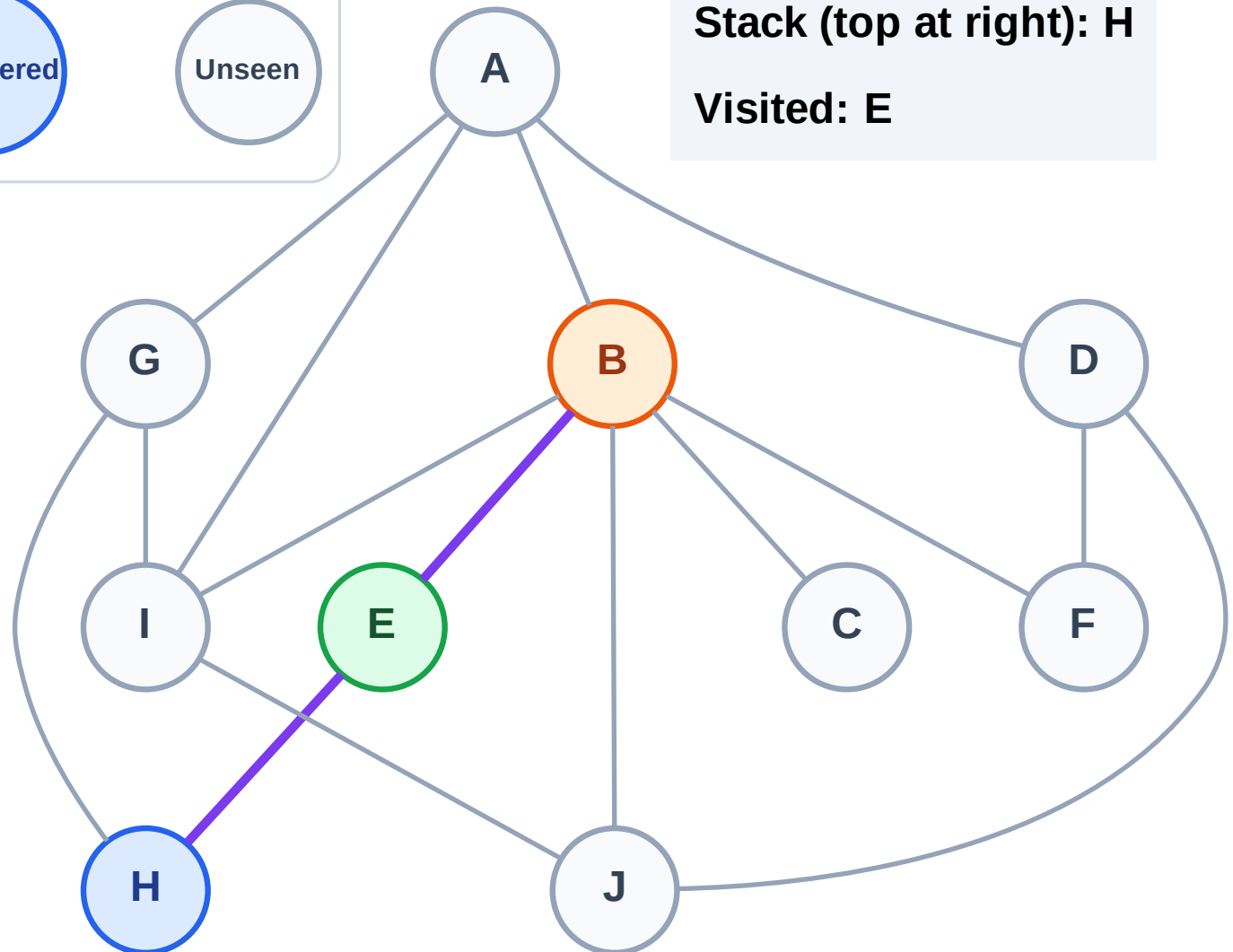
Processing

Discovered

Unseen

Stack (top at right): H

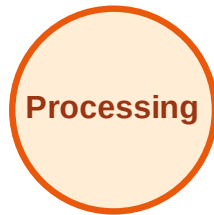
Visited: E



DFS Traversal

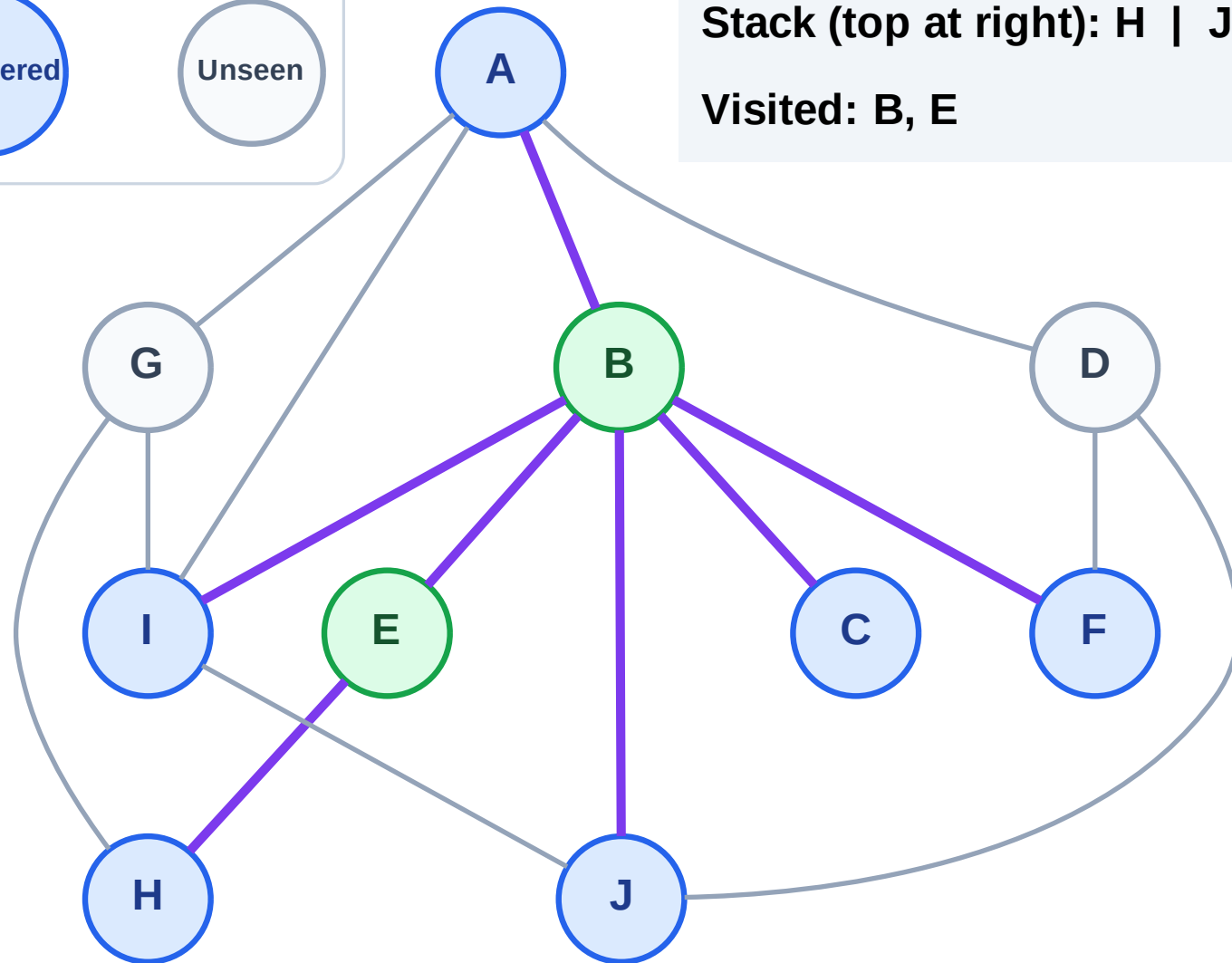
Step 5: Discover J, I, F, C, A from B

Legend



Stack (top at right): H | J | I | F | C | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

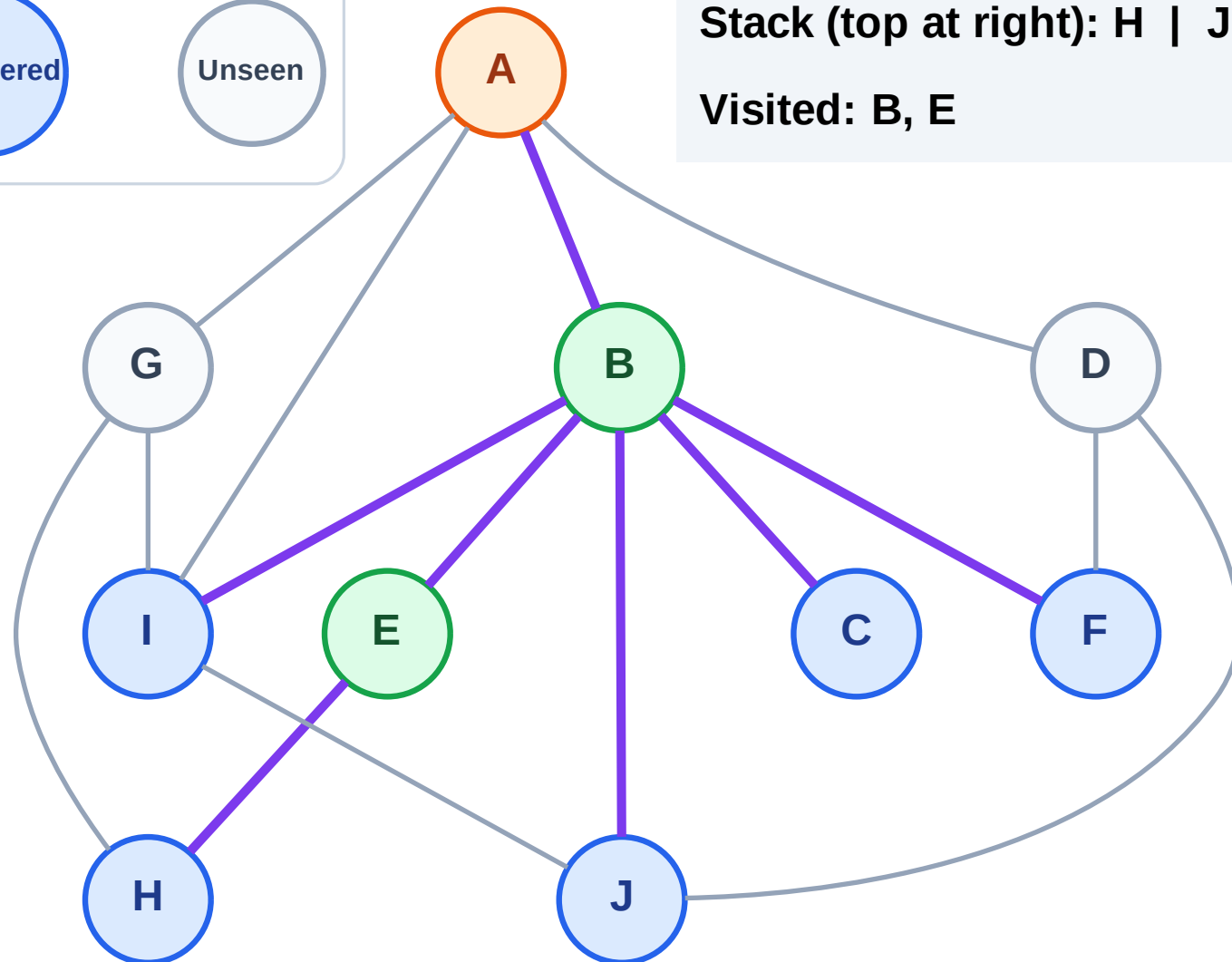
Processing

Discovered

Unseen

Stack (top at right): H | J | I | F | C

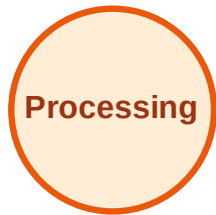
Visited: B, E



DFS Traversal

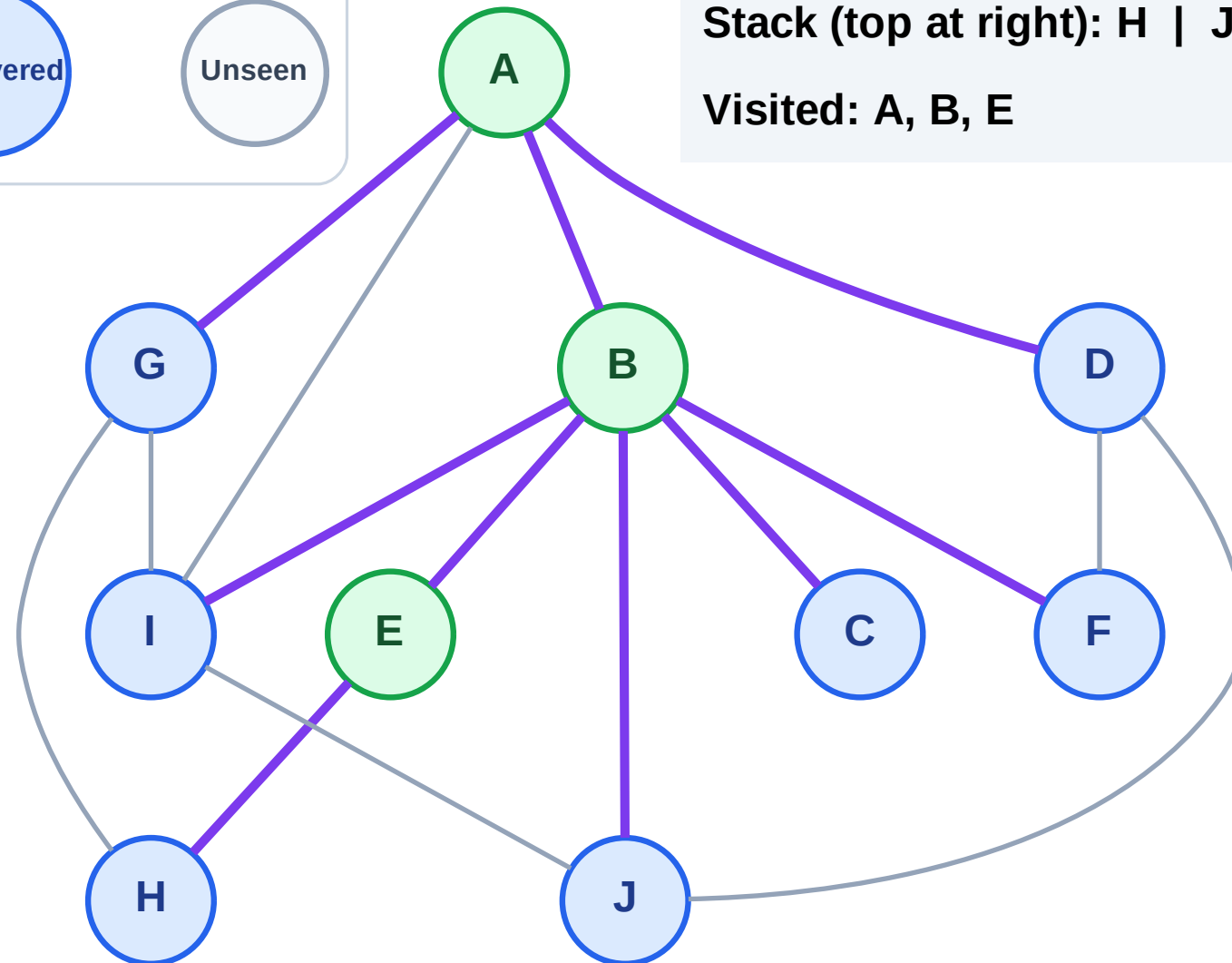
Step 7: Discover G, D from A

Legend



Stack (top at right): H | J | I | F | C | G | D

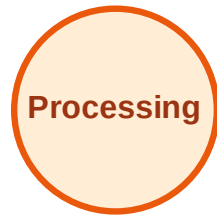
Visited: A, B, E



DFS Traversal

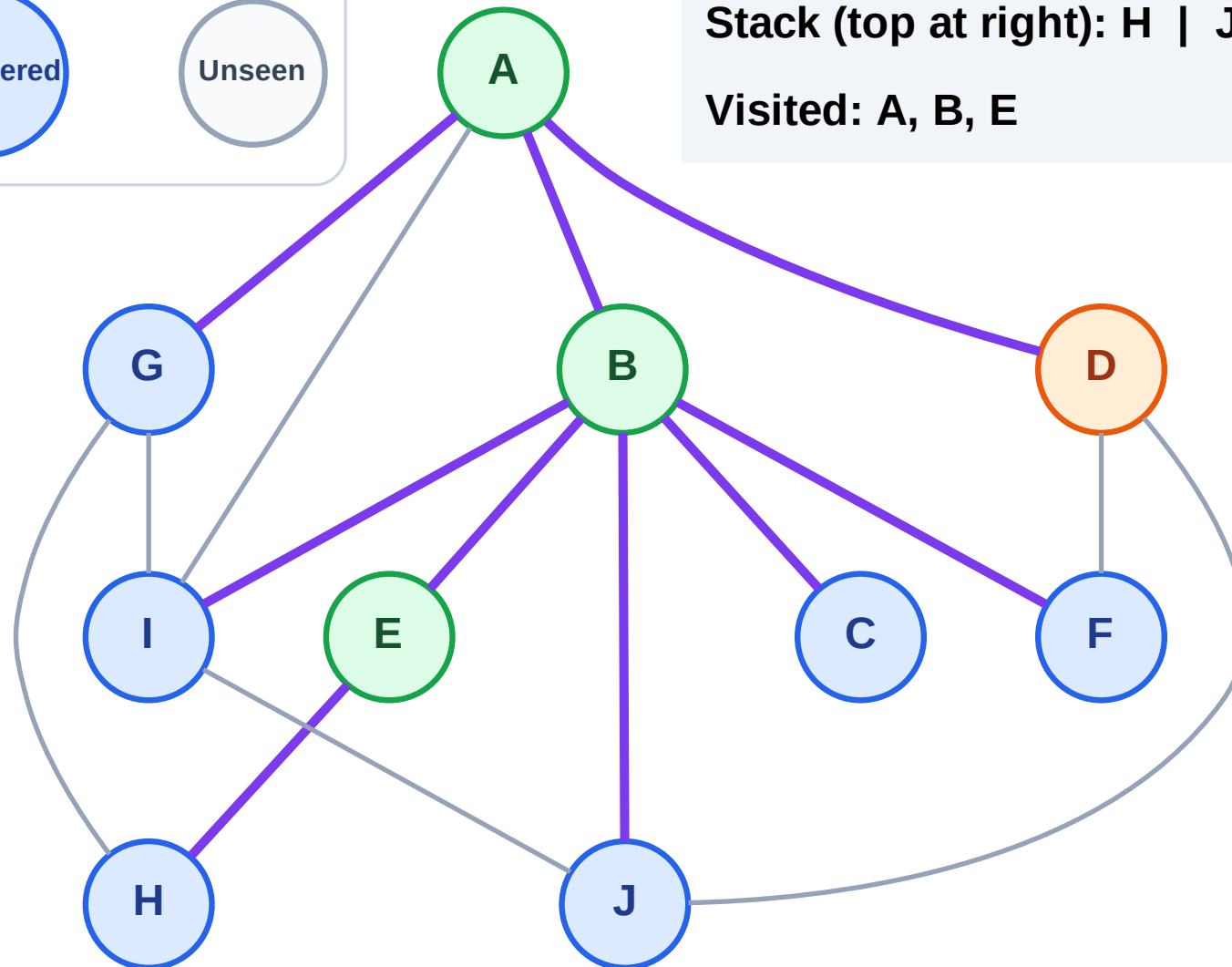
Step 8: Process node D

Legend



Stack (top at right): H | J | I | F | C | G

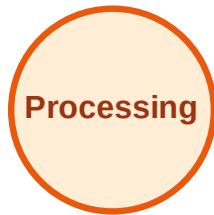
Visited: A, B, E



DFS Traversal

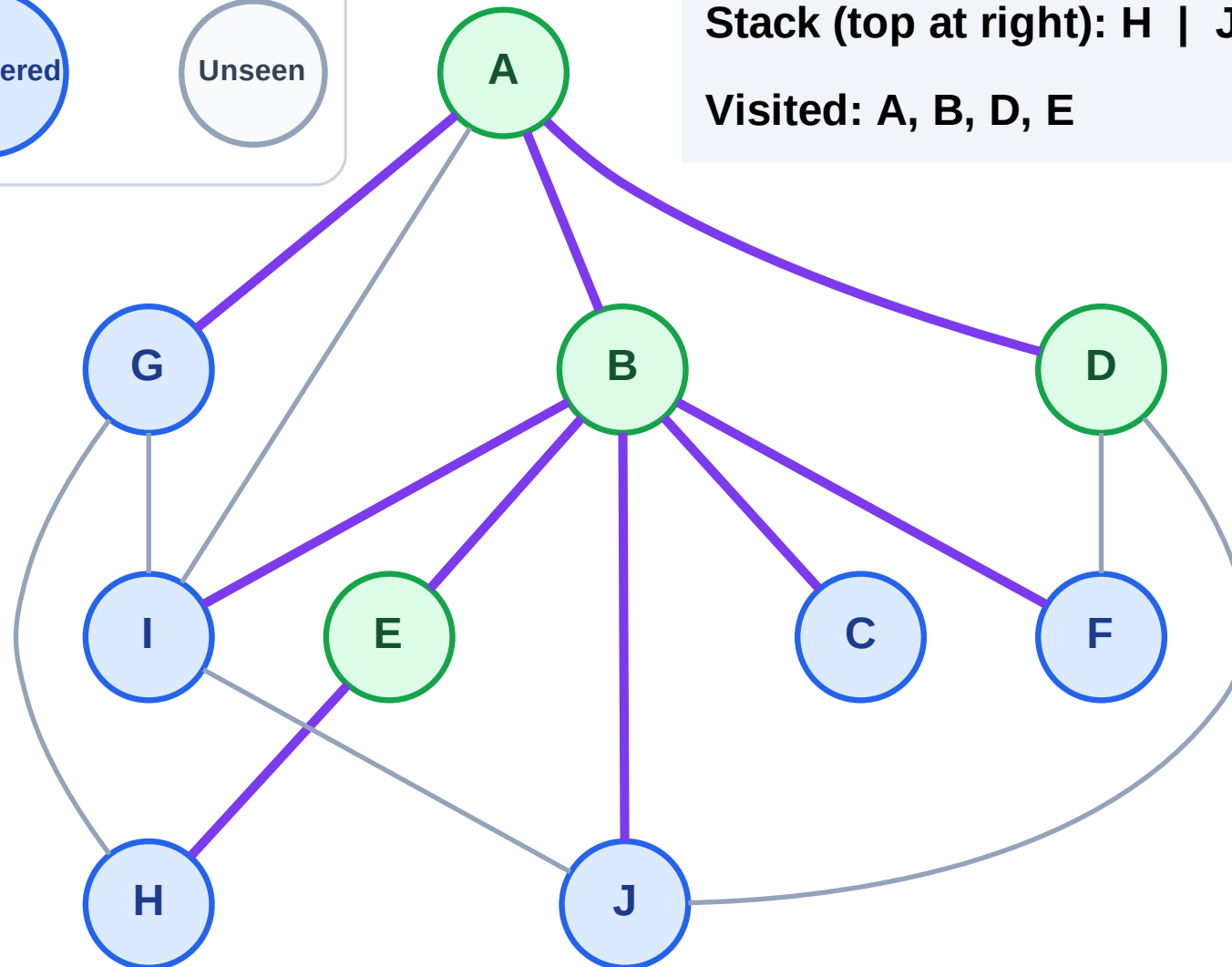
Step 9: No new neighbors from D

Legend



Stack (top at right): H | J | I | F | C | G

Visited: A, B, D, E



DFS Traversal

Step 10: Process node G

Legend

Complete

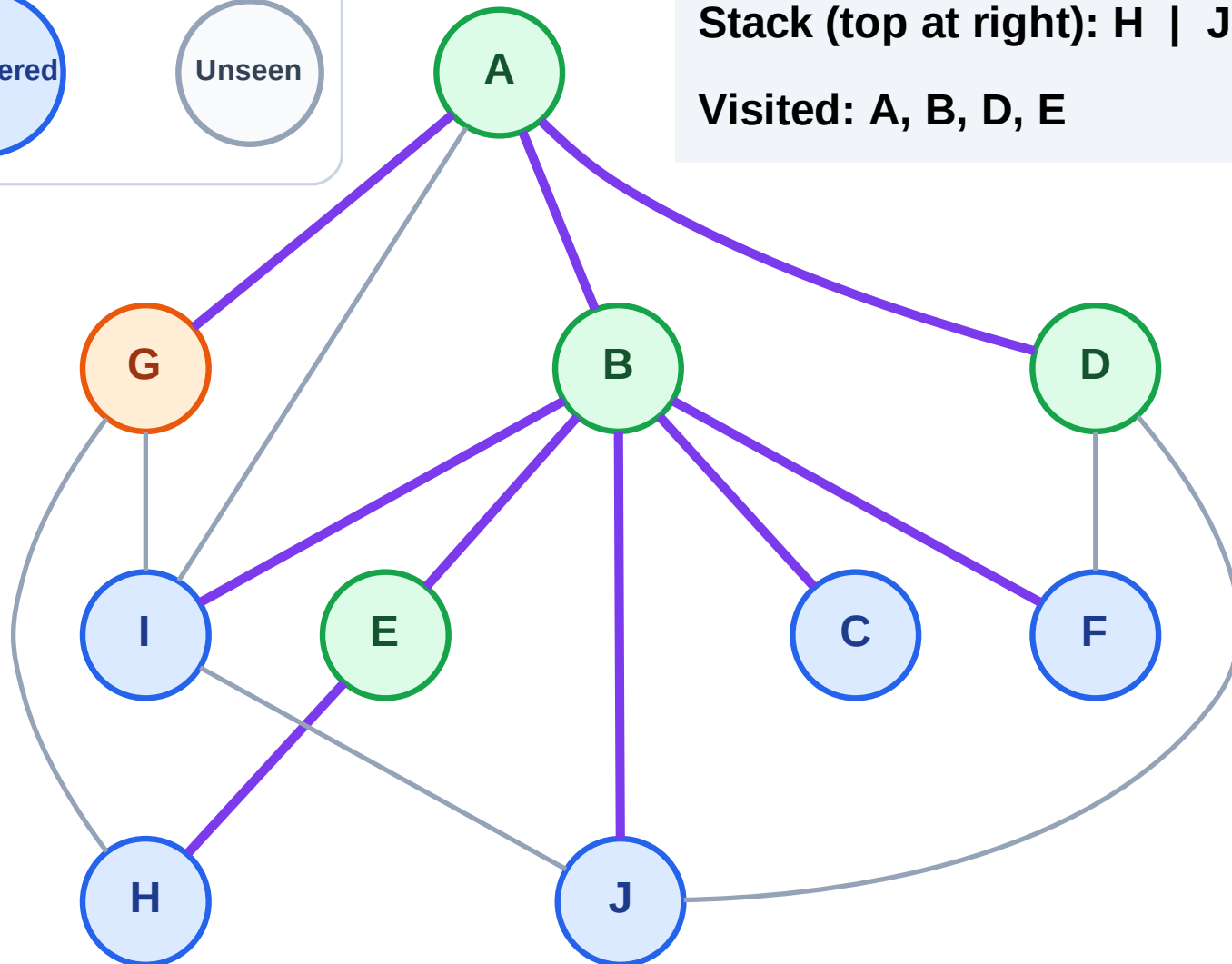
Processing

Discovered

Unseen

Stack (top at right): H | J | I | F | C

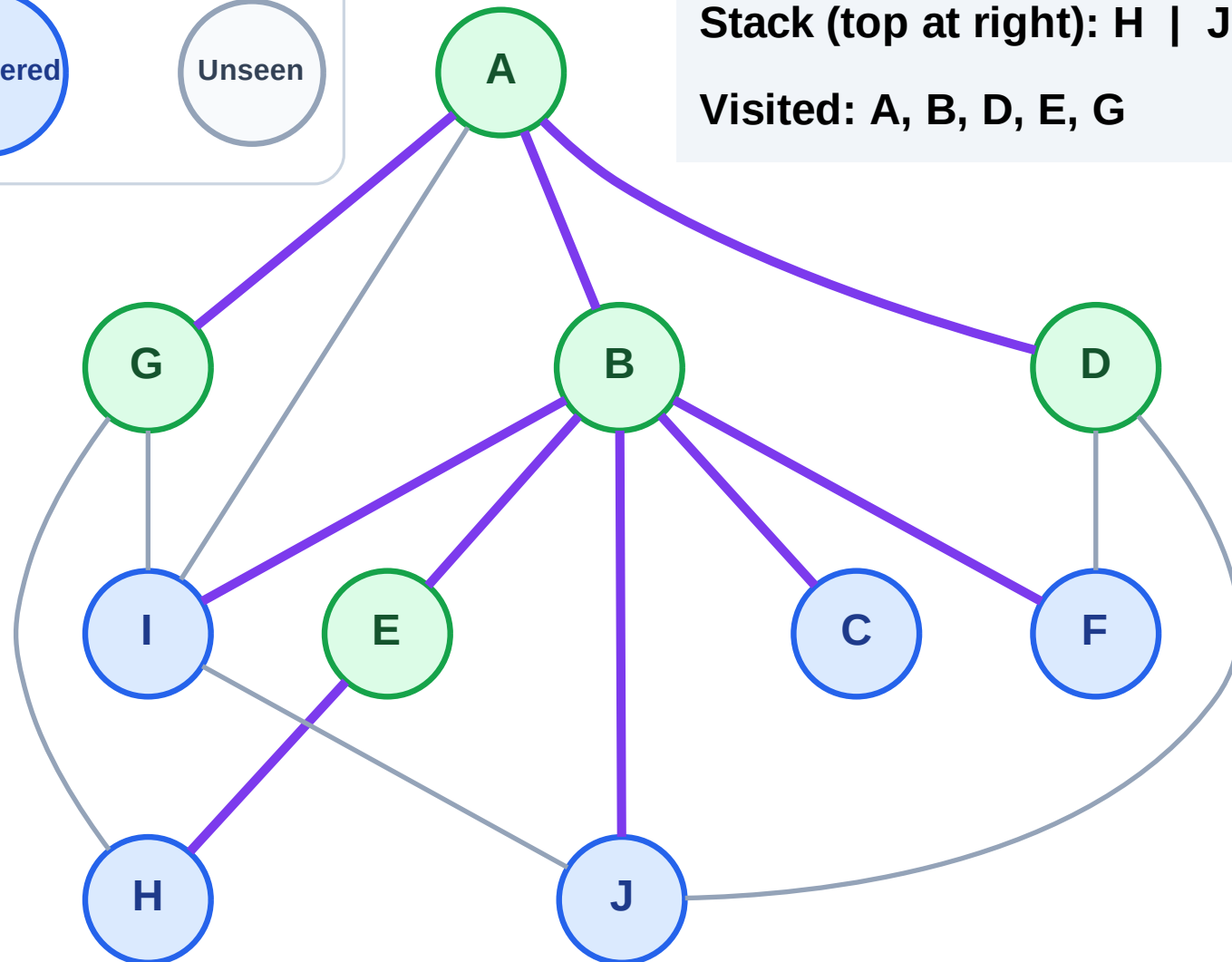
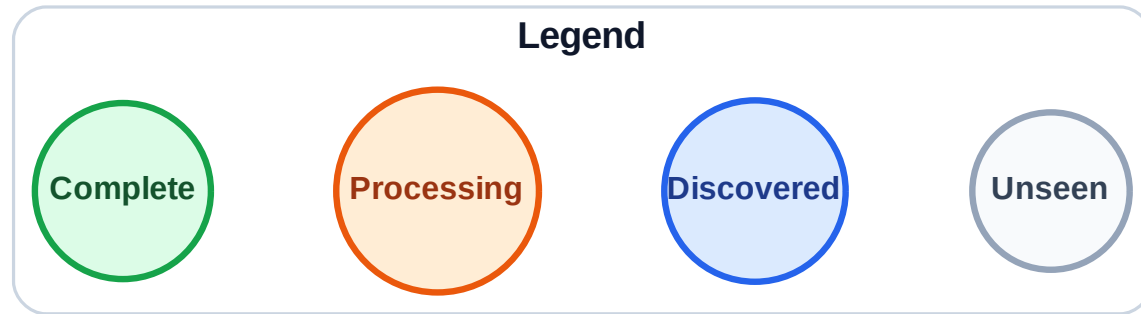
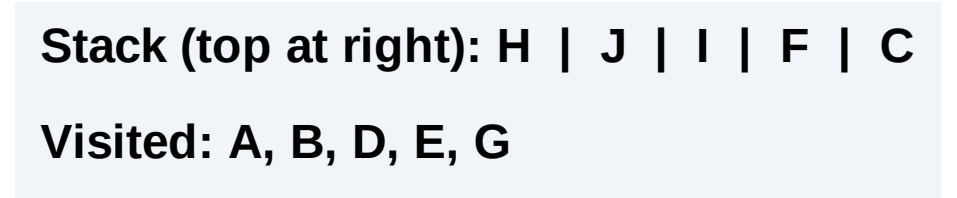
Visited: A, B, D, E



DFS Traversal

Step 11: No new neighbors from G

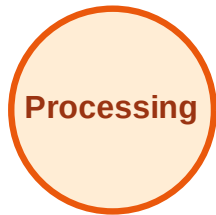
Step 11: No new neighbors from G



DFS Traversal

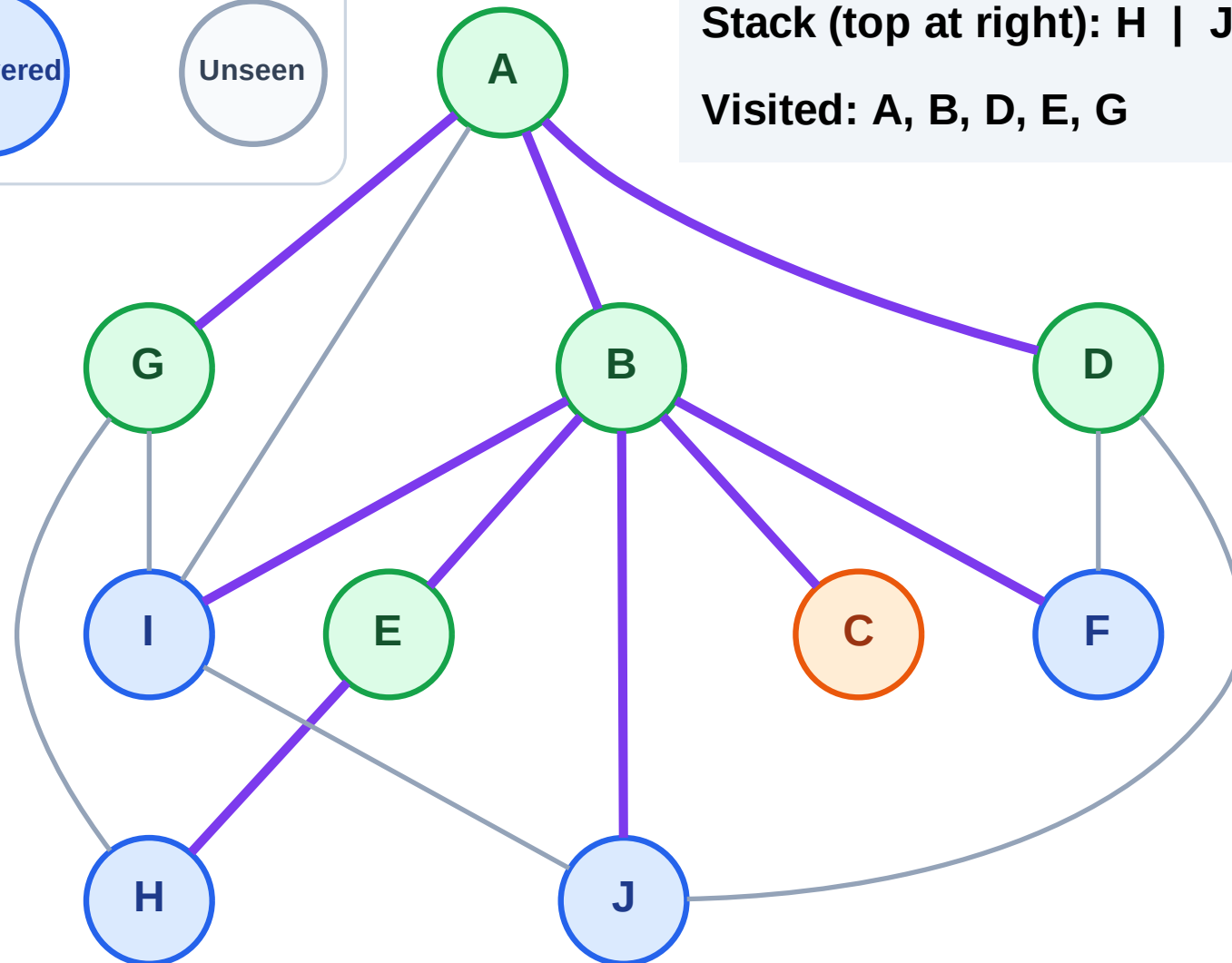
Step 12: Process node C

Legend



Stack (top at right): H | J | I | F

Visited: A, B, D, E, G



DFS Traversal

Step 13: No new neighbors from C

Legend

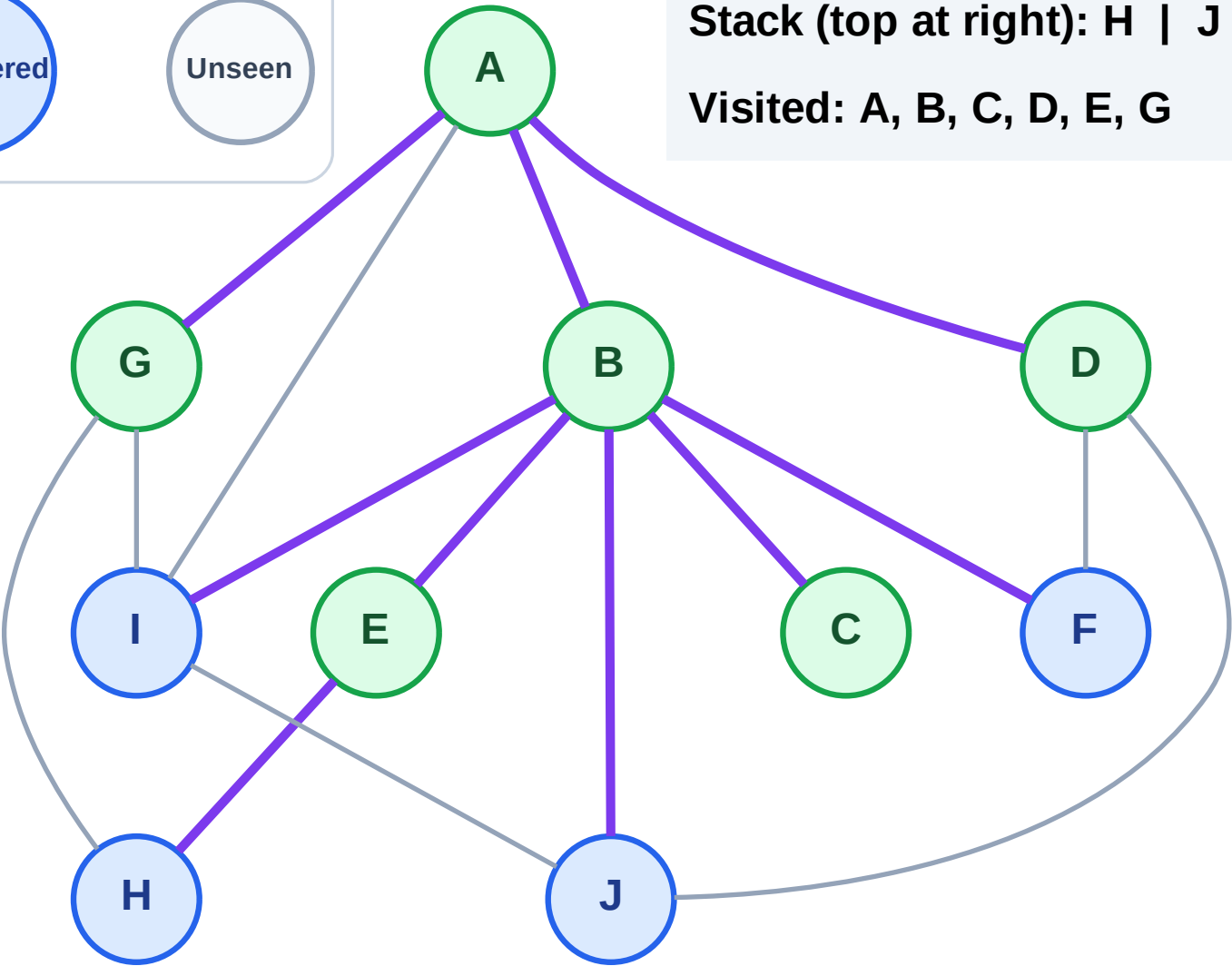
Complete

Processing

Discovered

Unseen

Stack (top at right): H | J | I | F
Visited: A, B, C, D, E, G



DFS Traversal

Step 14: Process node F

Legend

Complete

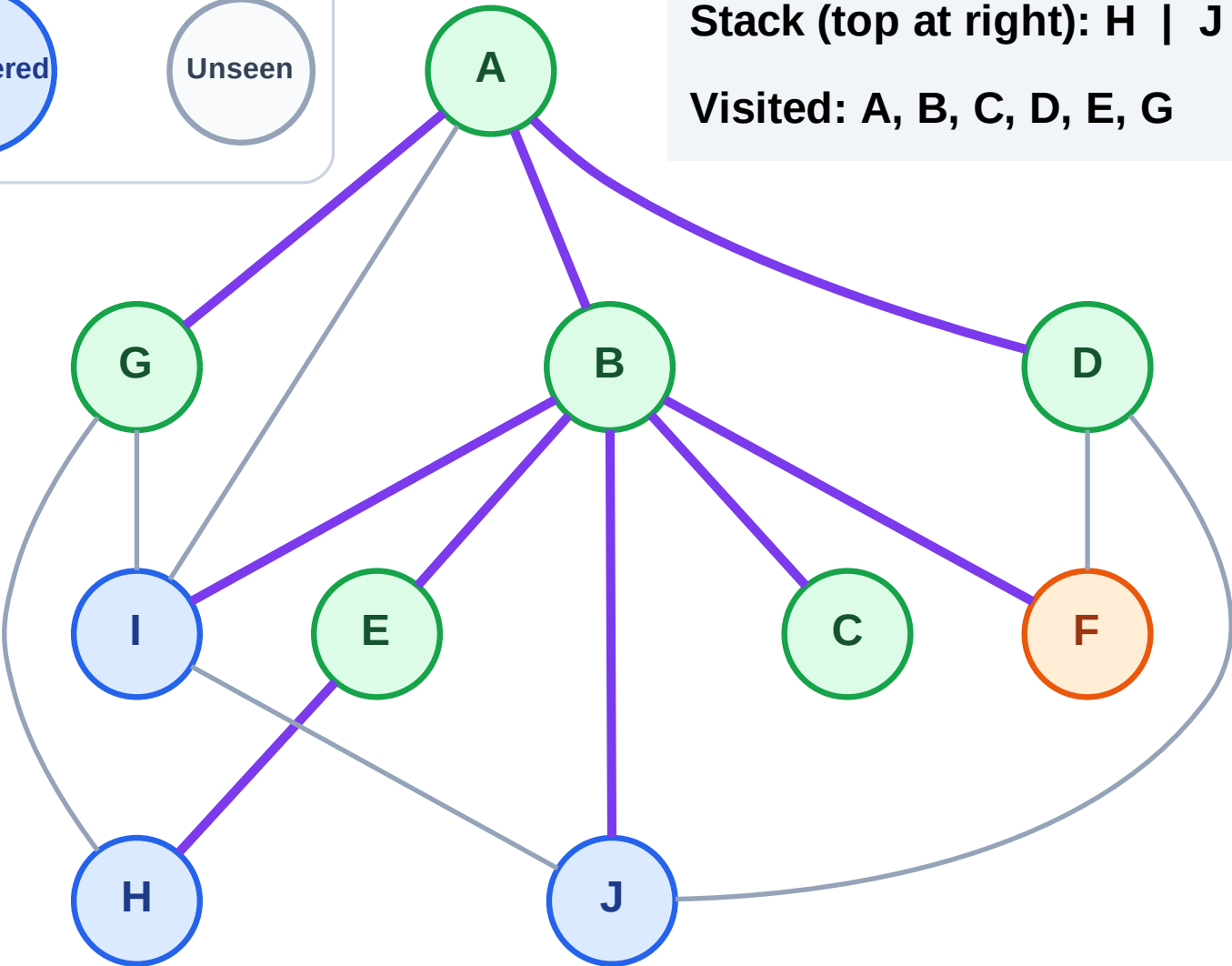
Processing

Discovered

Unseen

Stack (top at right): H | J | I

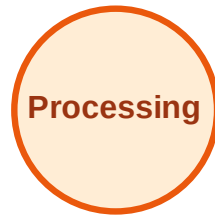
Visited: A, B, C, D, E, G



DFS Traversal

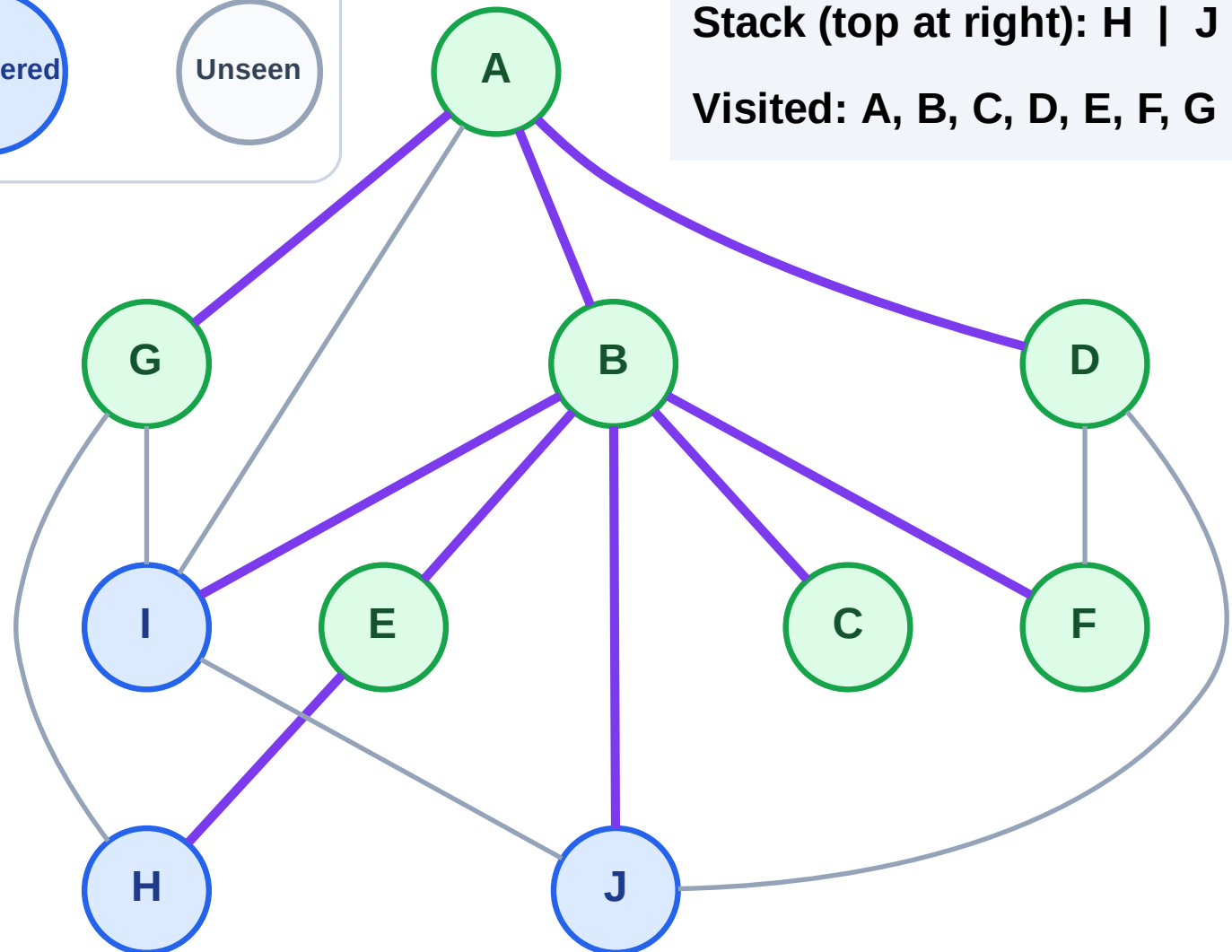
Step 15: No new neighbors from F

Legend



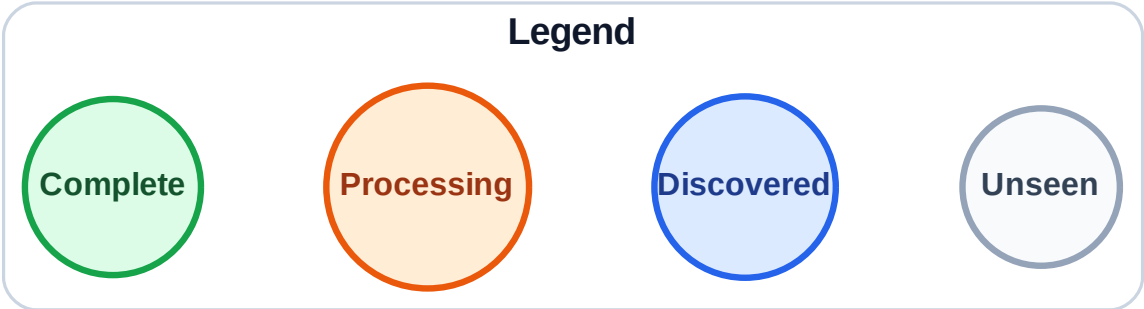
Stack (top at right): H | J | I

Visited: A, B, C, D, E, F, G

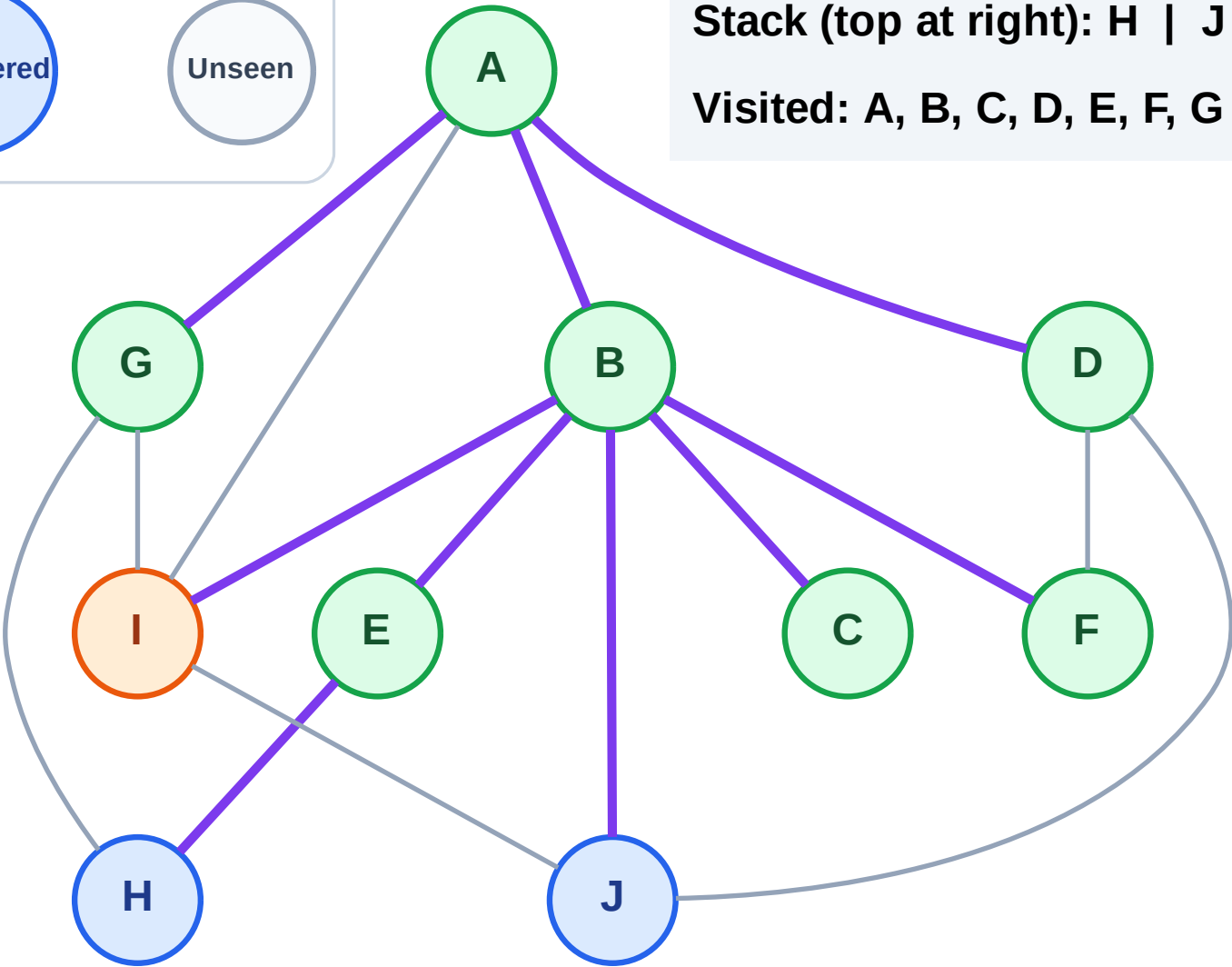


DFS Traversal

Step 16: Process node I



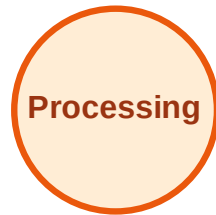
Stack (top at right): H | J
Visited: A, B, C, D, E, F, G



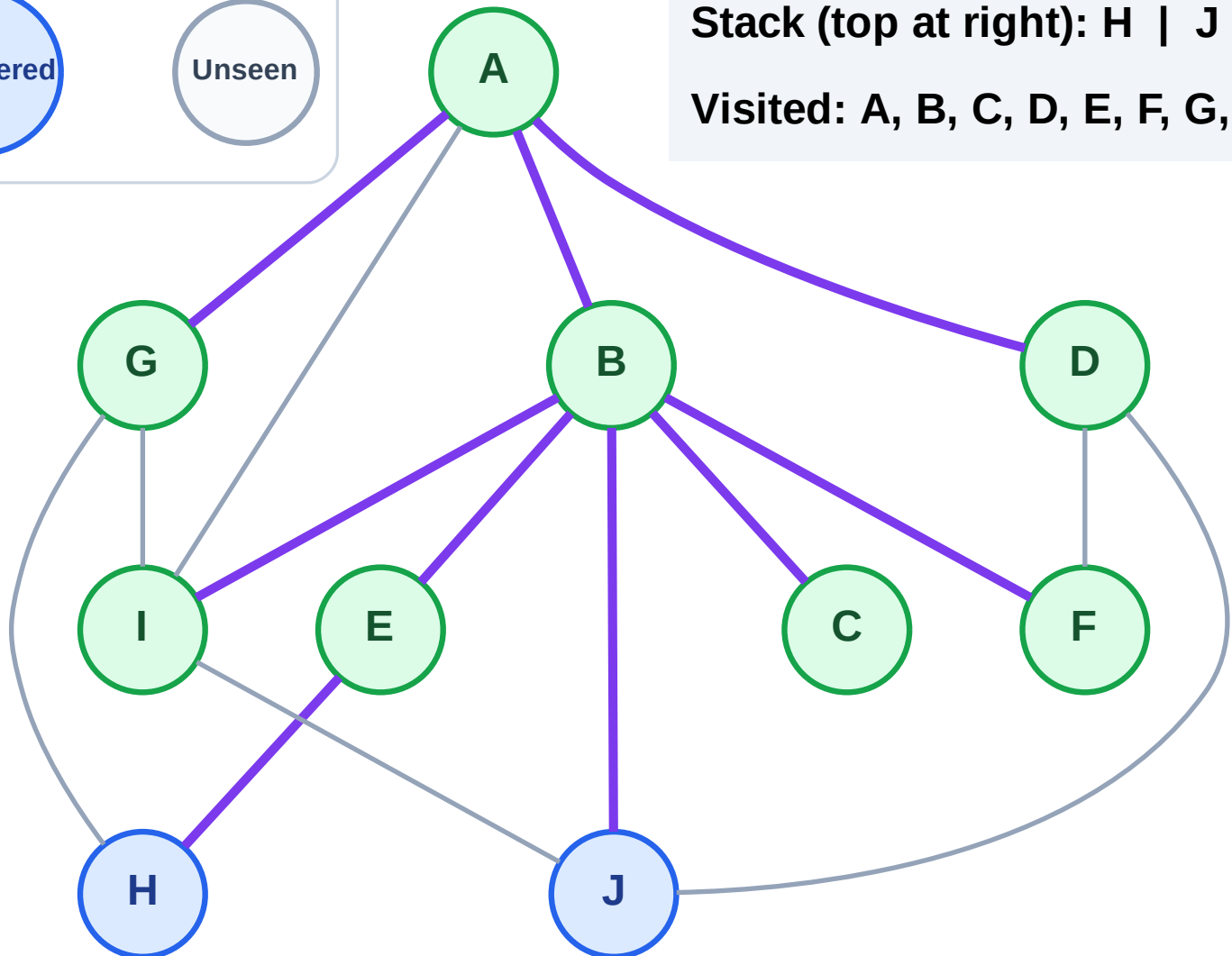
DFS Traversal

Step 17: No new neighbors from I

Legend

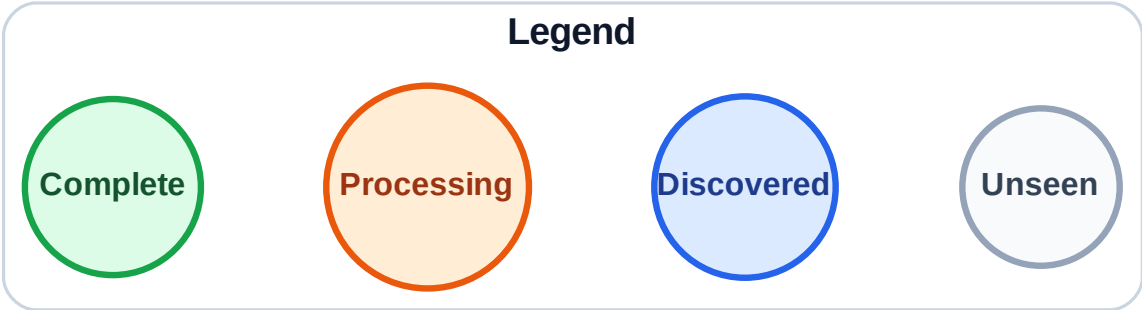


Stack (top at right): H | J
Visited: A, B, C, D, E, F, G, I

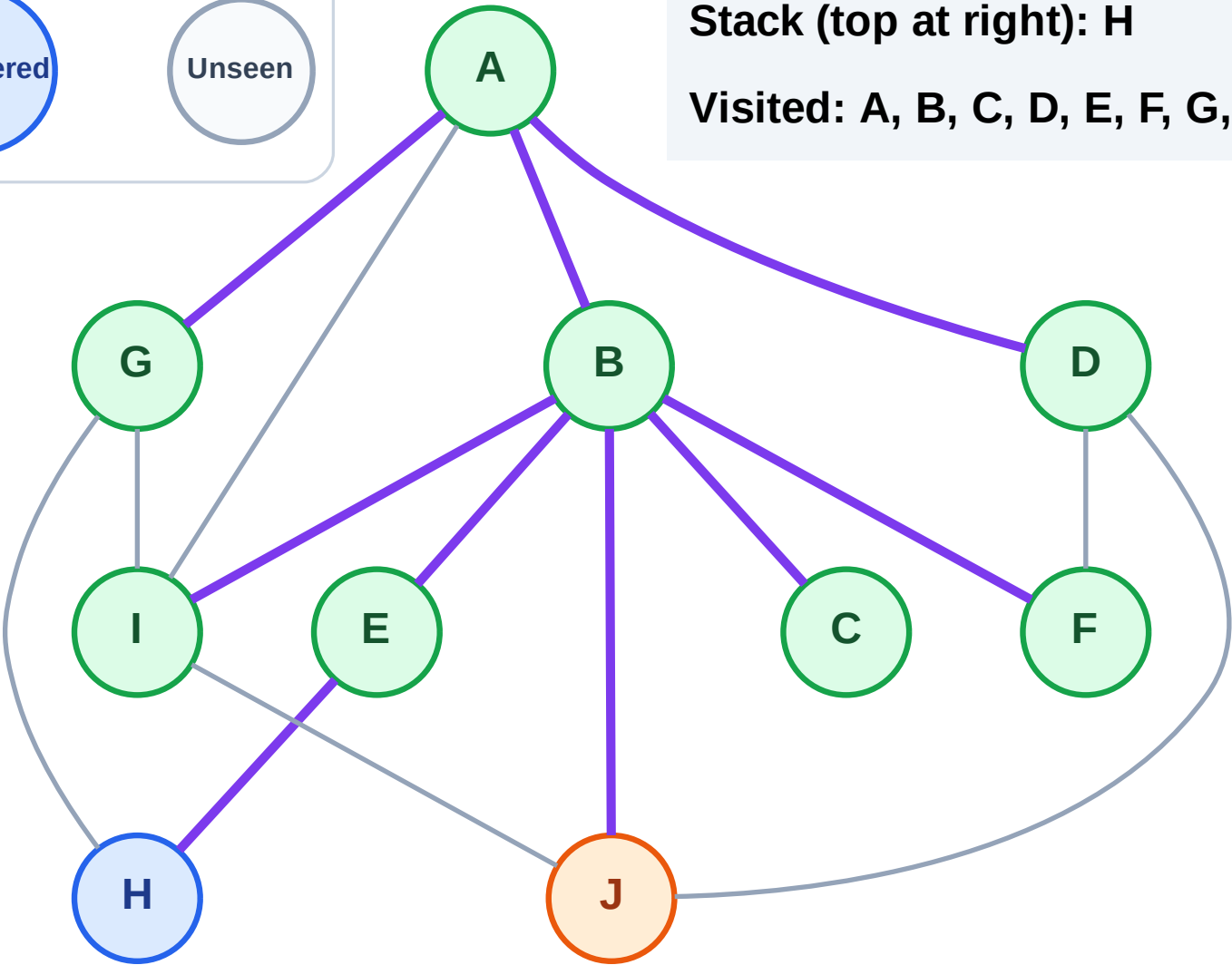


DFS Traversal

Step 18: Process node J



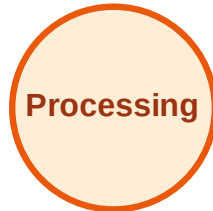
Stack (top at right): H
Visited: A, B, C, D, E, F, G, I



DFS Traversal

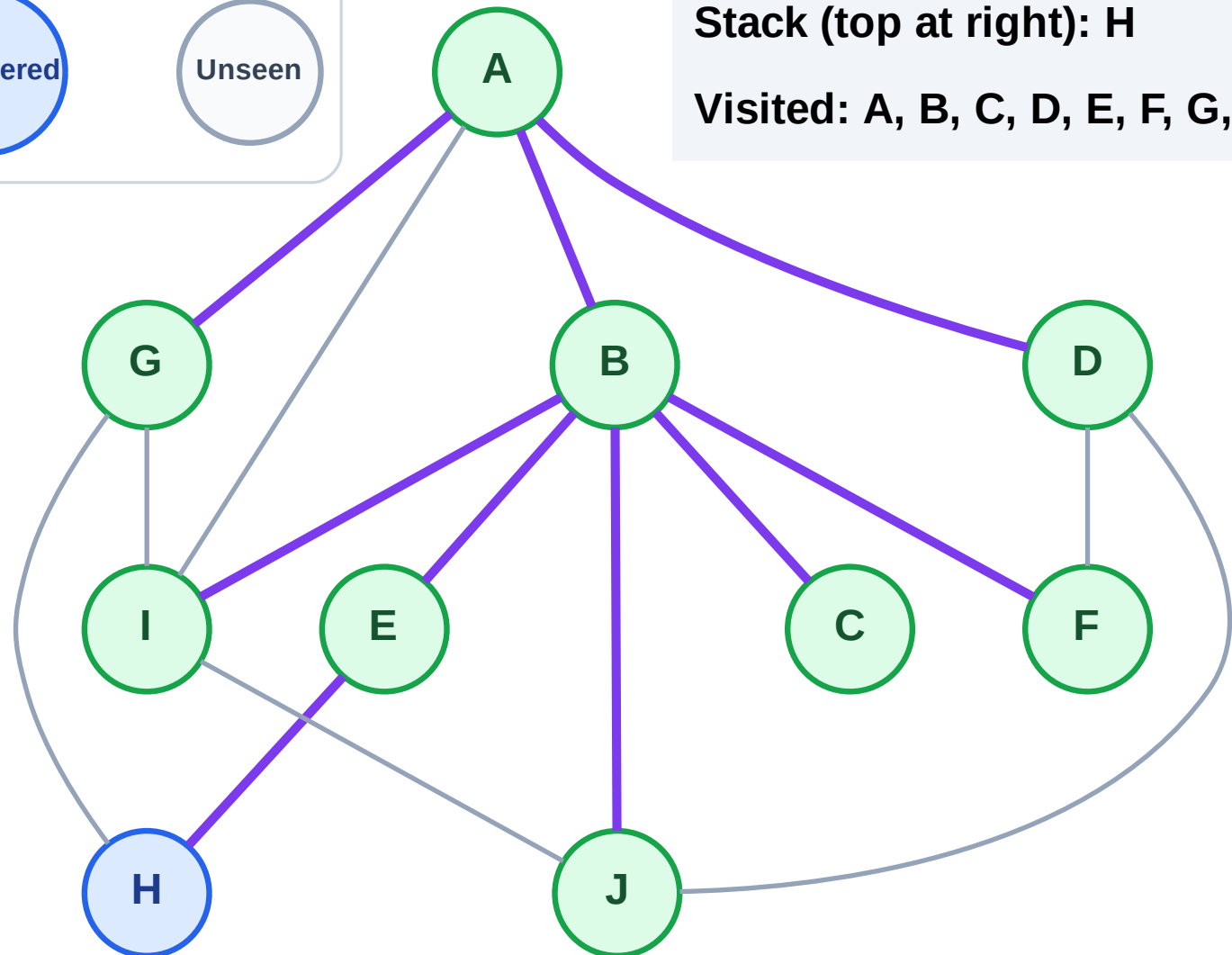
Step 19: No new neighbors from J

Legend



Stack (top at right): H

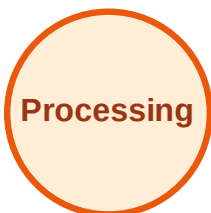
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

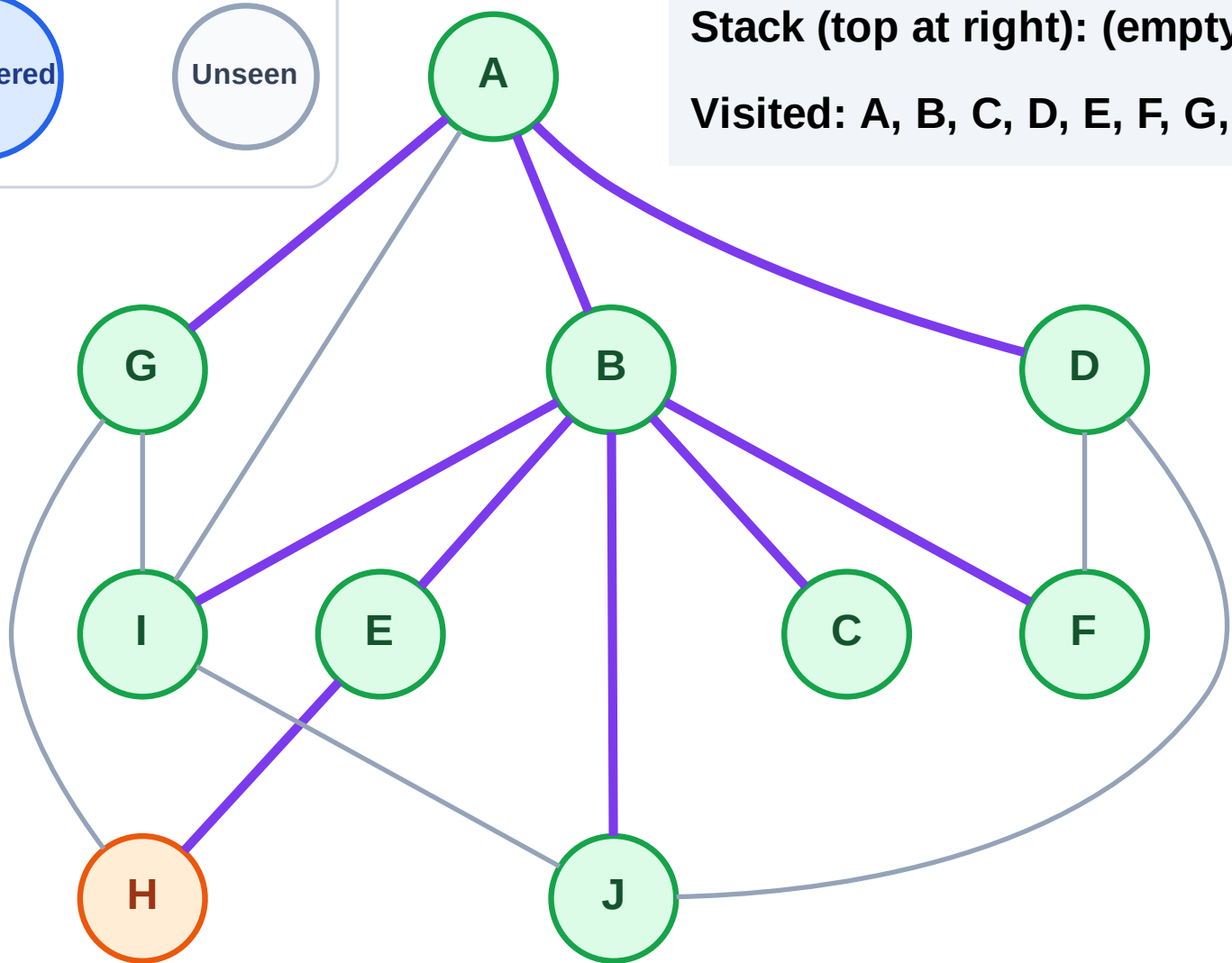
Step 20: Process node H

Legend



Stack (top at right): (empty)

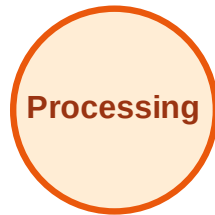
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

